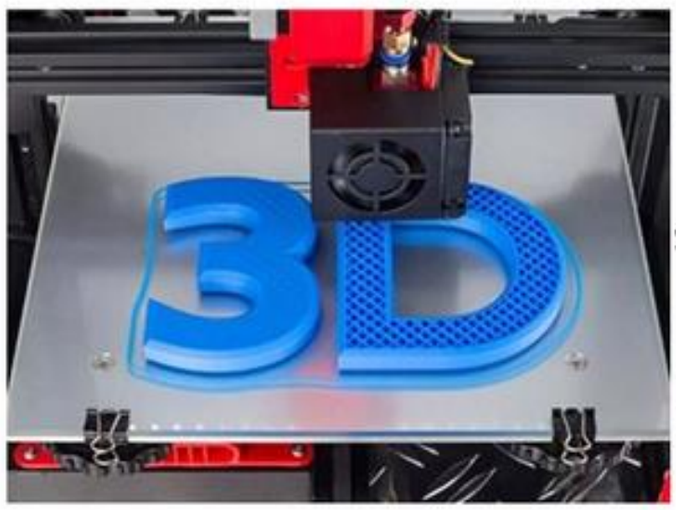
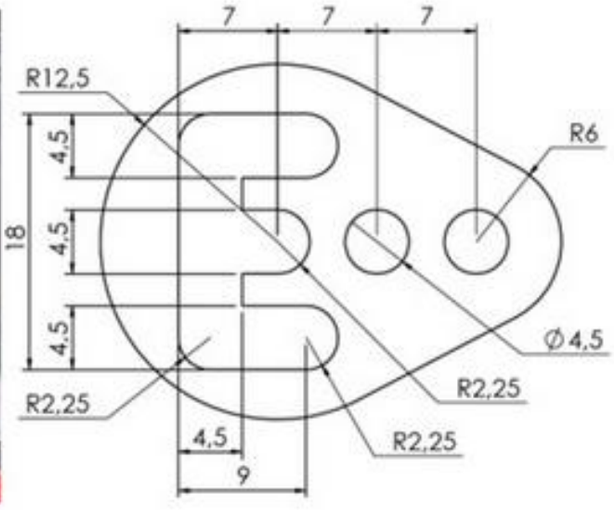
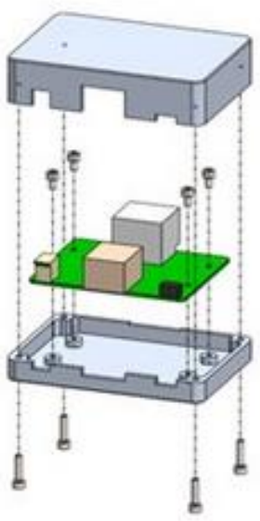
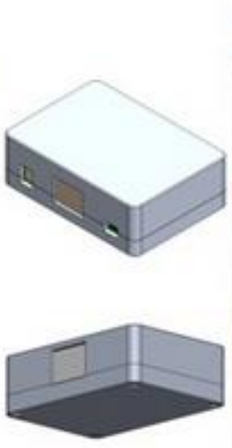
UNIVERSIDAD DE MÁLAGA | ESCUELA DE INGENIERÍAS INDUSTRIALES | roboRESCUE UMA | L I N K BY UMA-ATECH | K Skills

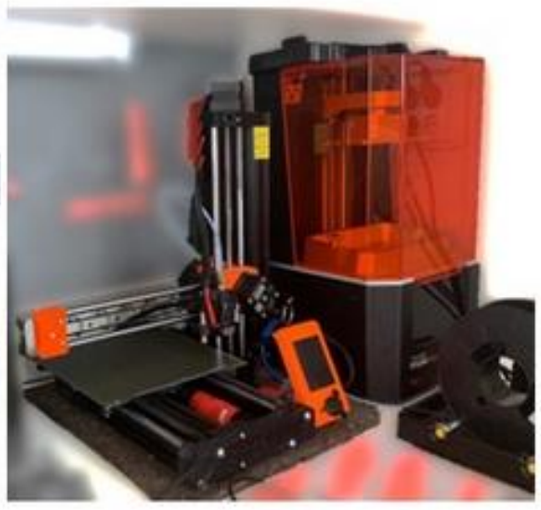
Diseño con SolidWorks para **Impresión 3D**







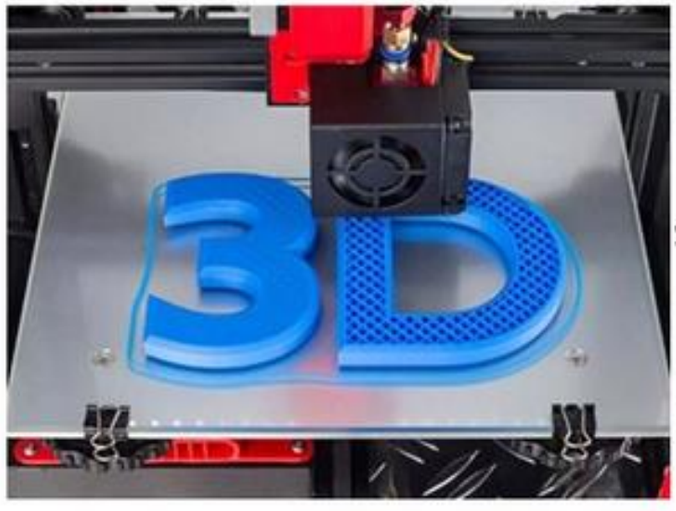


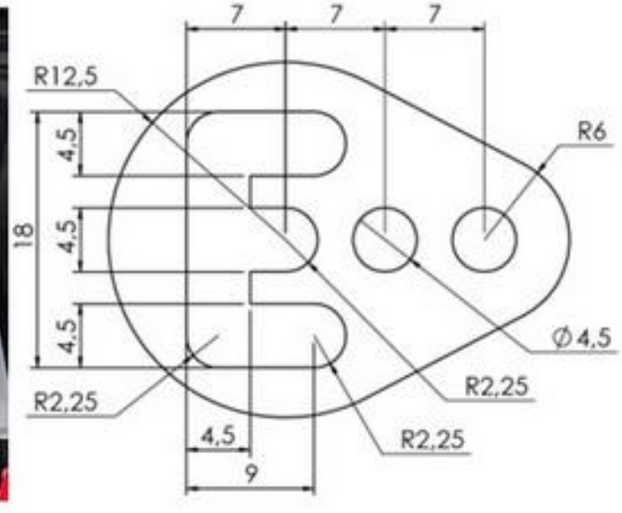


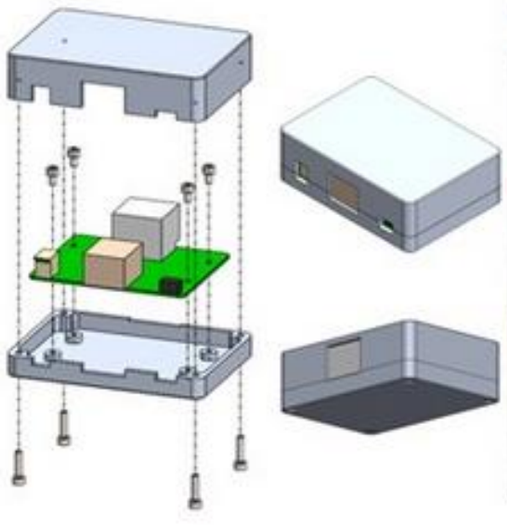
CLASE 1 - B

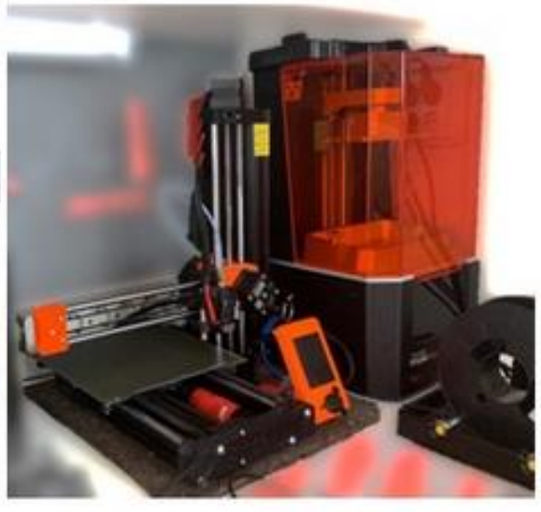
Programa de laminación

Uso general









IMPRESIÓN 3D – **LAMINADOR** (SLICER)

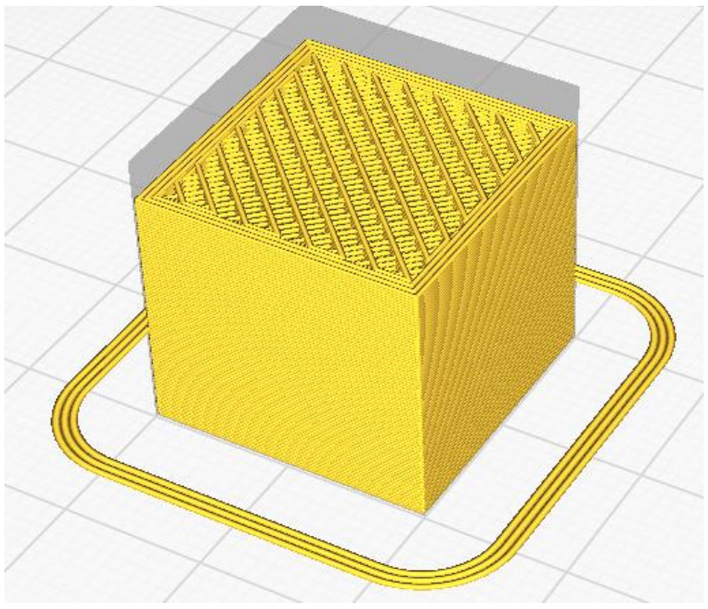


MODELO LAMINADO

.STL

Laminador (Slicer)

cura.



IMPRESIÓN 3D – **CURA** (SLICER)

UltiMaker Impresoras 3D **Software** Materiales Ecosystem Aplicaciones Saber más Soporte Solicitar presupuesto

Ultimaker › Software › **UltiMaker Cura: software de impresió...**

UltiMaker Cura

Elegido por millones de usuarios, UltiMaker Cura es el software de impresión 3D más popular del mundo. Prepare impresiones con unos cuantos clics, intégrelo con el software de CAD para un mejor flujo de trabajo o aproveche las configuraciones personalizadas para conseguir un control más completo.

Descarga gratuita ¿Tienes una UltiMaker?



Buscar versiones anteriores

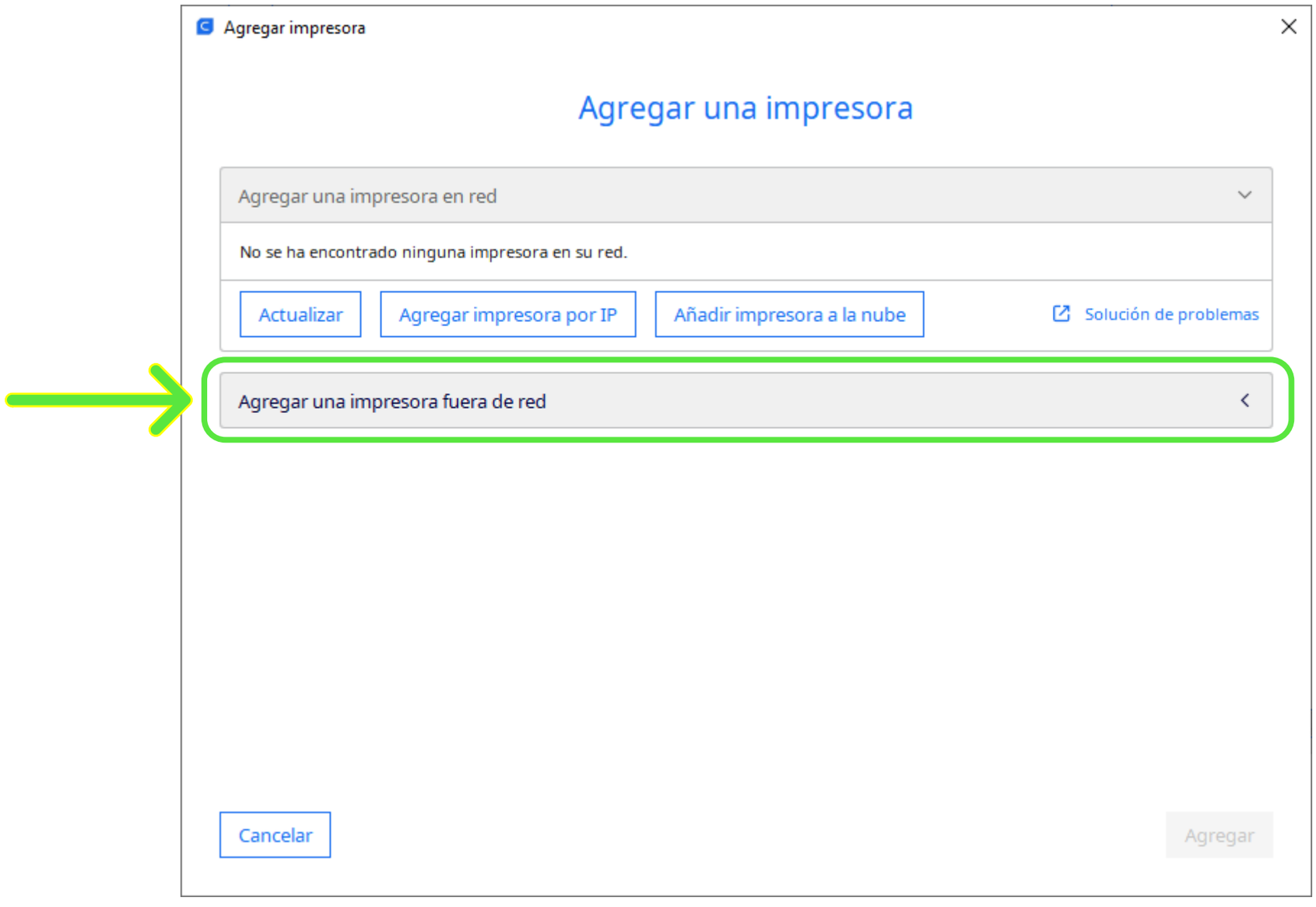
Requisitos del sistema

Ultimaker Cura 5.2.2
Windows, 64 bit

https://ultimaker.com/es/software/ultimaker-cura

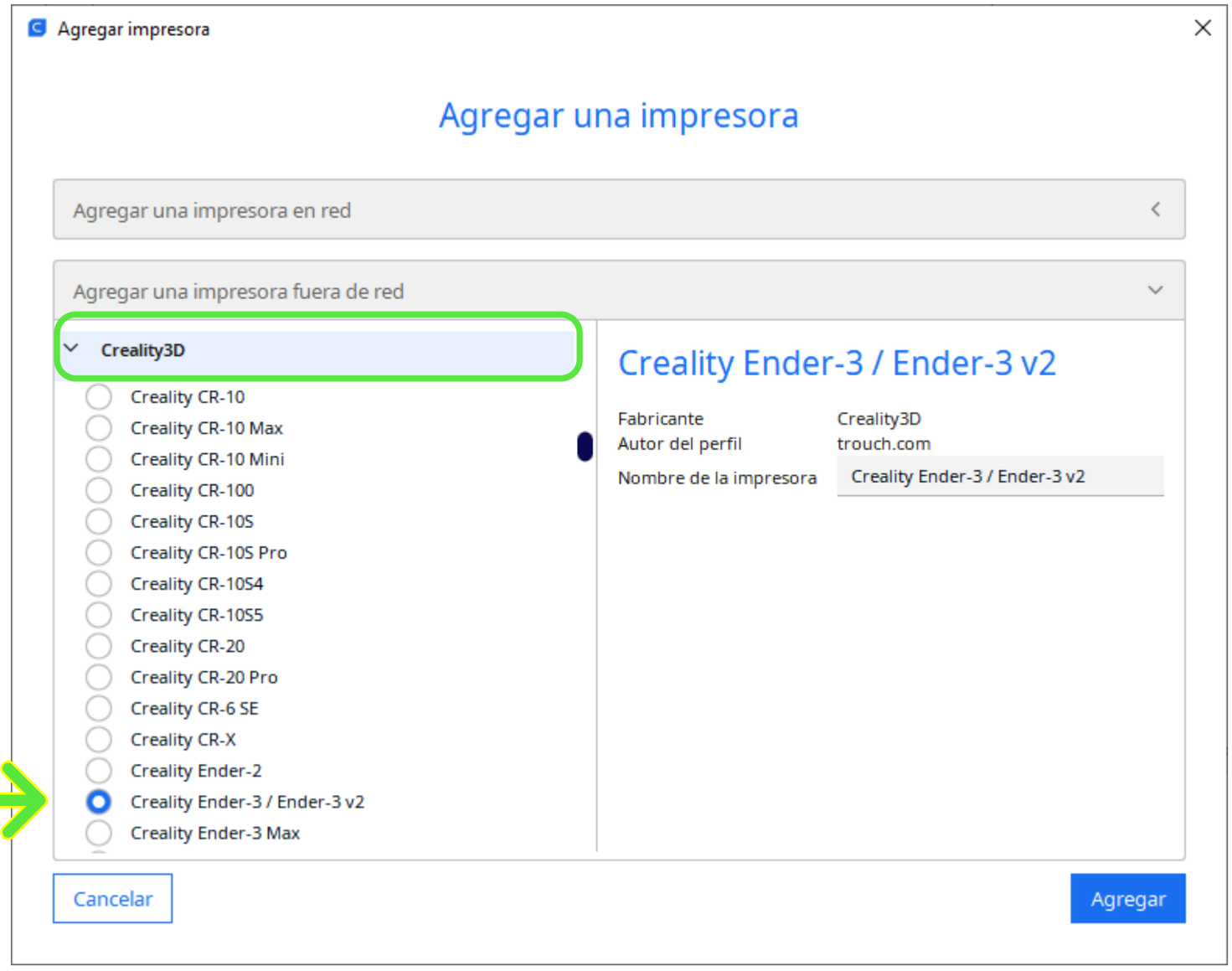
CURA– Configurar impresora

1 IMPRESORA SIN CONEXIÓN A RED



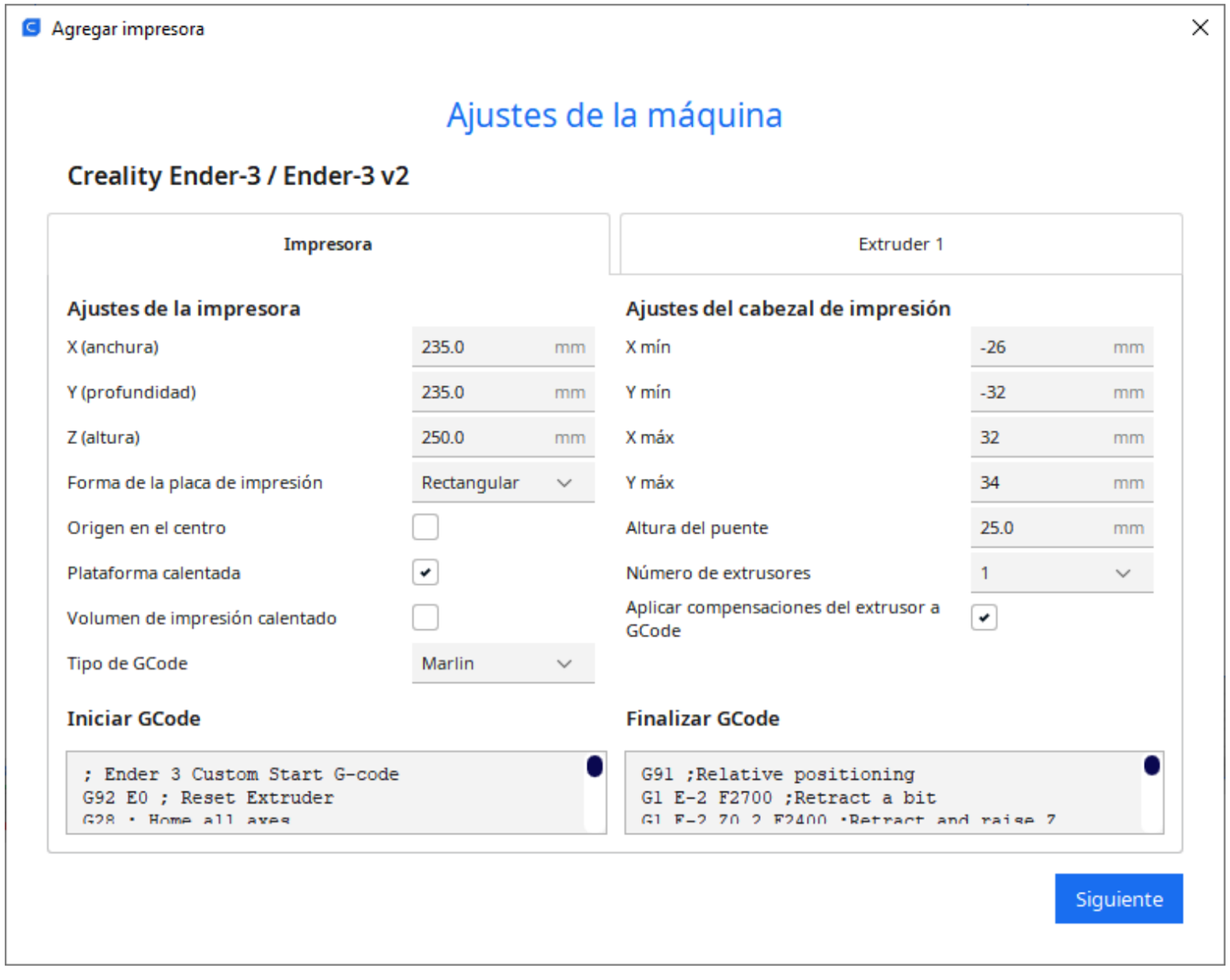
CURA– Configurar impresora

2 SELECCIONAR IMPRESORA 3D

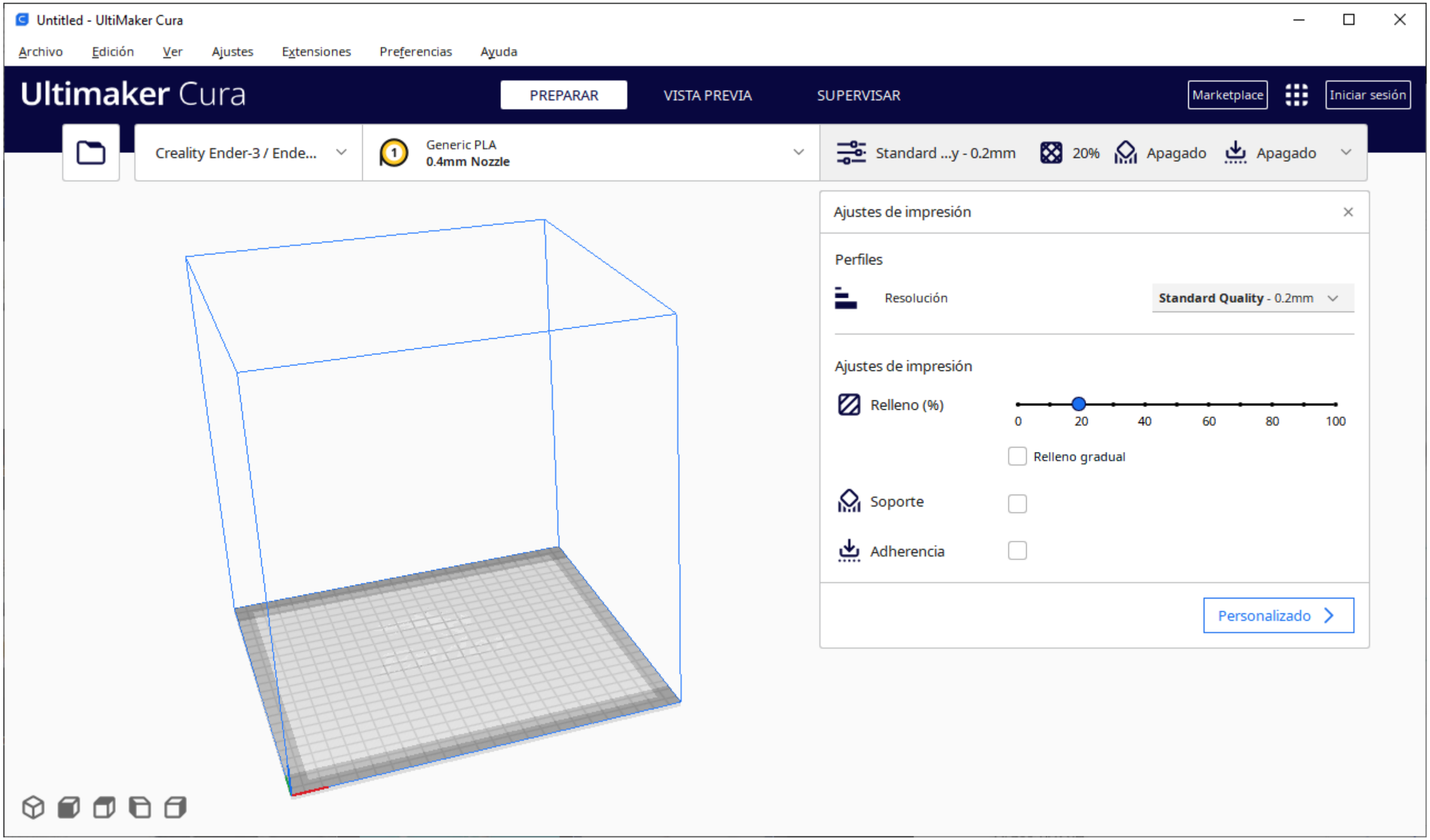


CURA– Configurar impresora

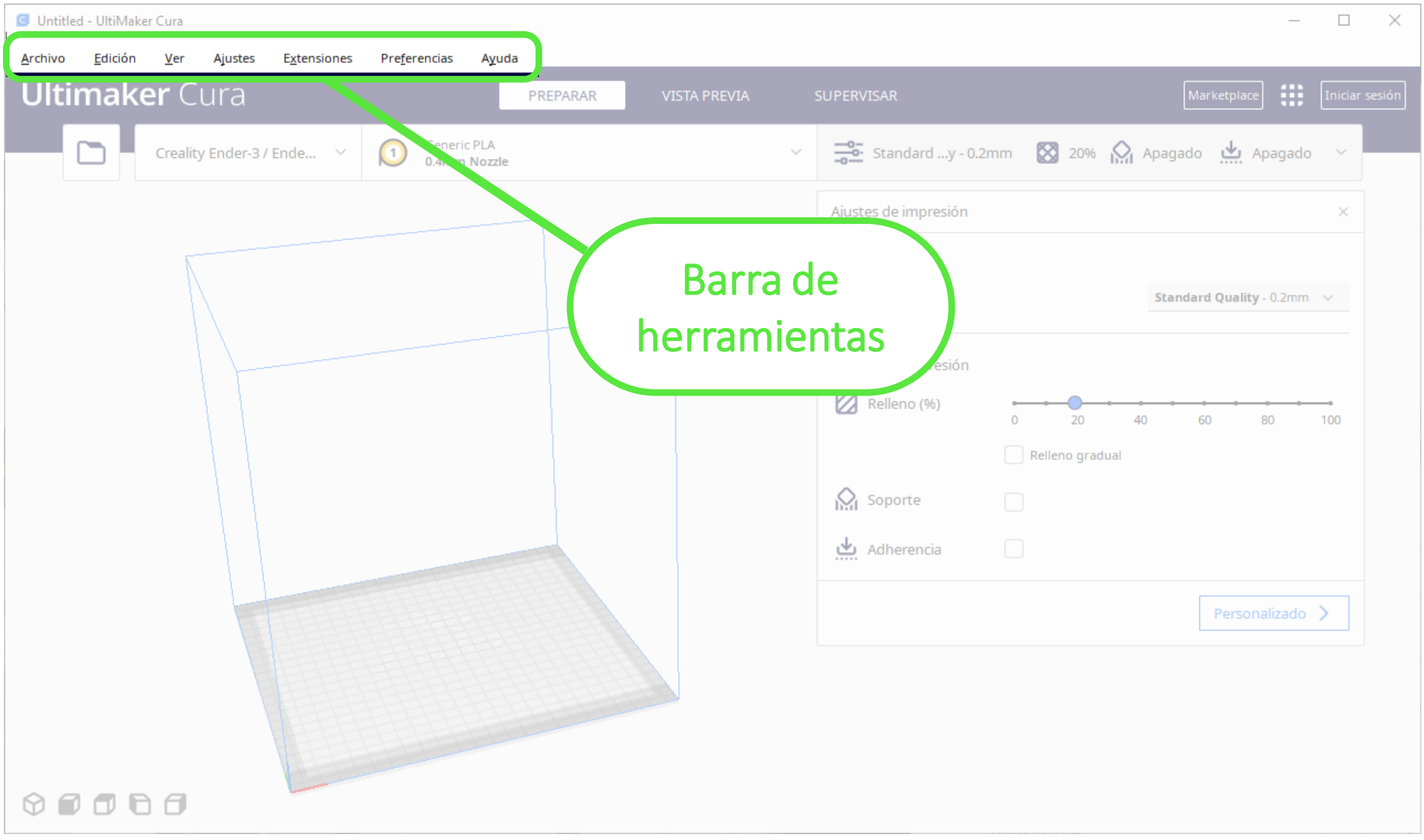
3 AJUSTES DE IMPRESORA 3D



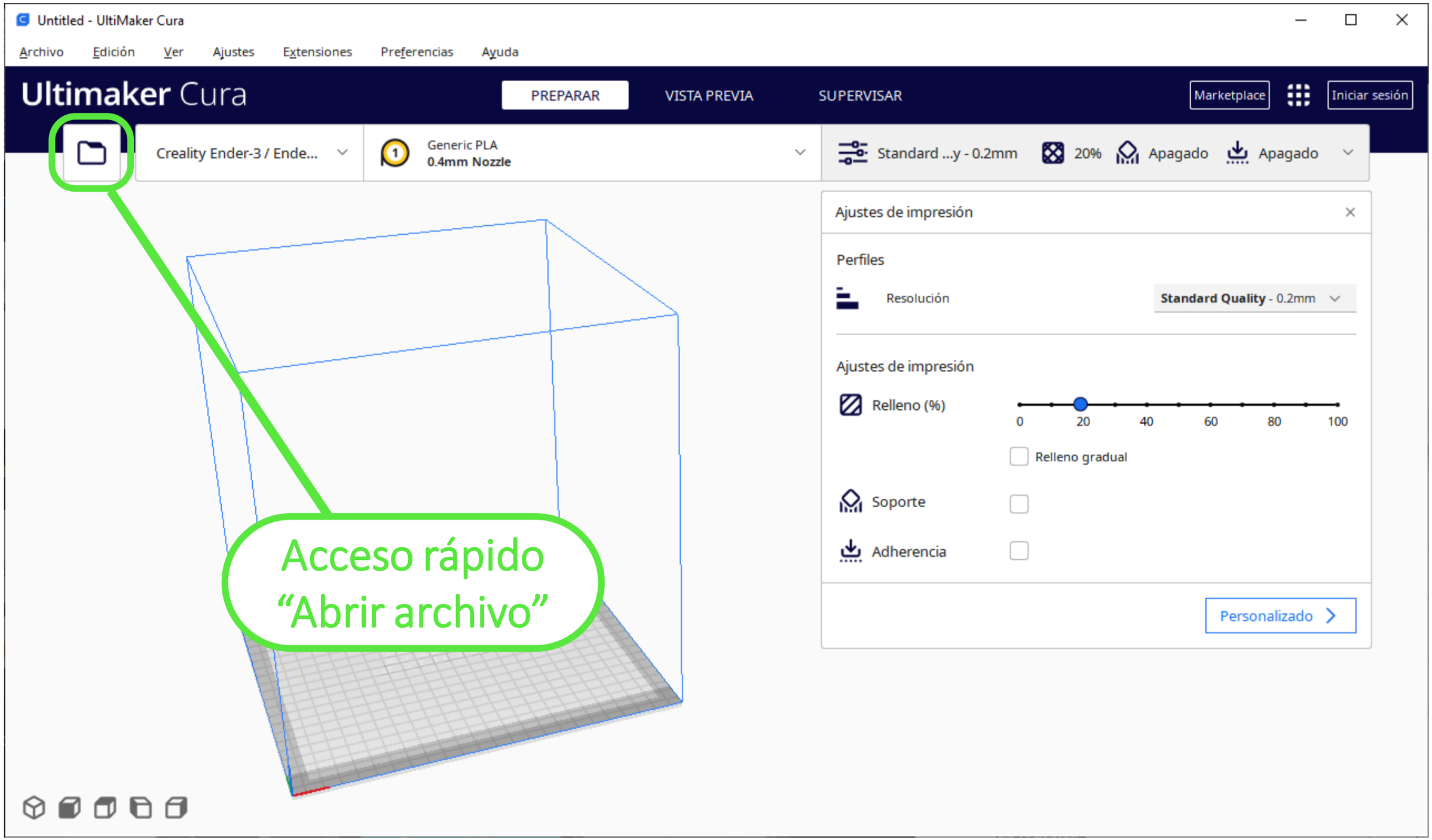
CURA– **Entorno** de trabajo



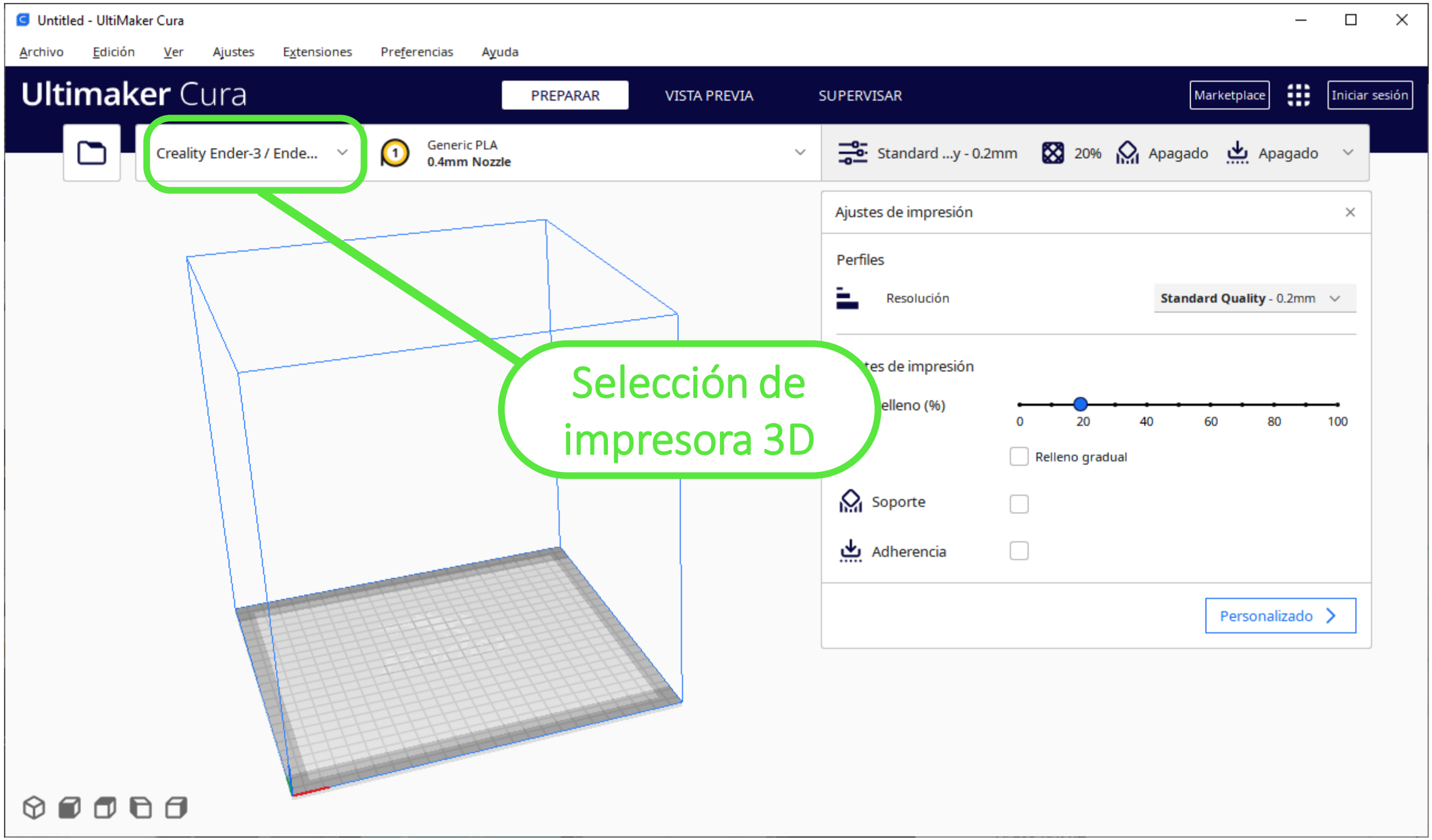
CURA– **Entorno** de trabajo



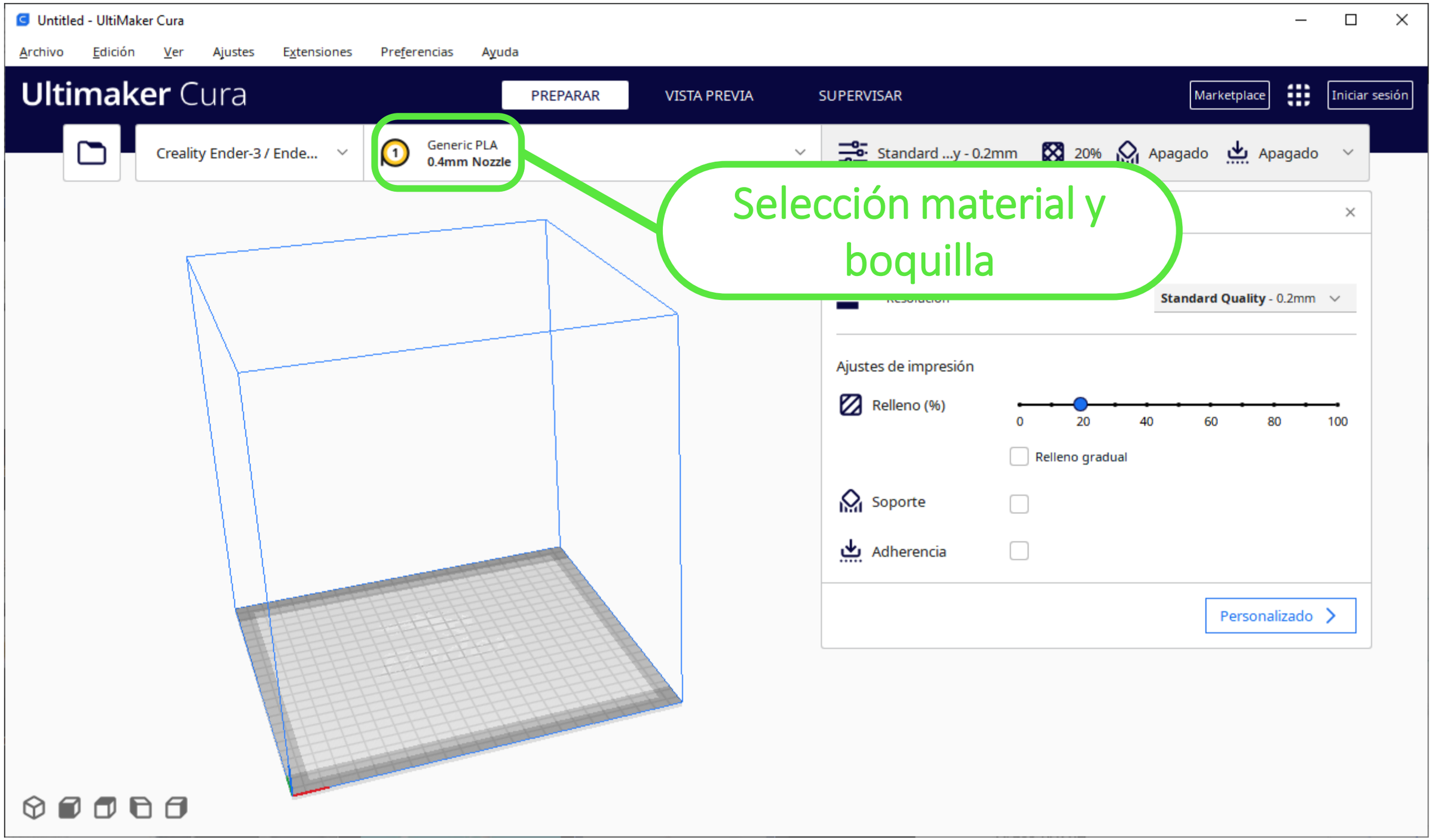
CURA– **Entorno** de trabajo



CURA– **Entorno** de trabajo



CURA– **Entorno** de trabajo

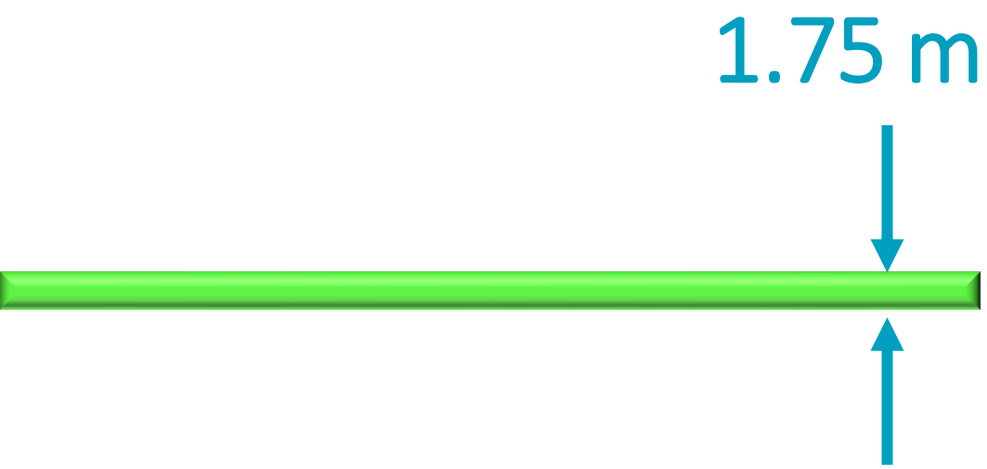


FDM – **Material** de impresión

BOBINA DE FILAMENTO





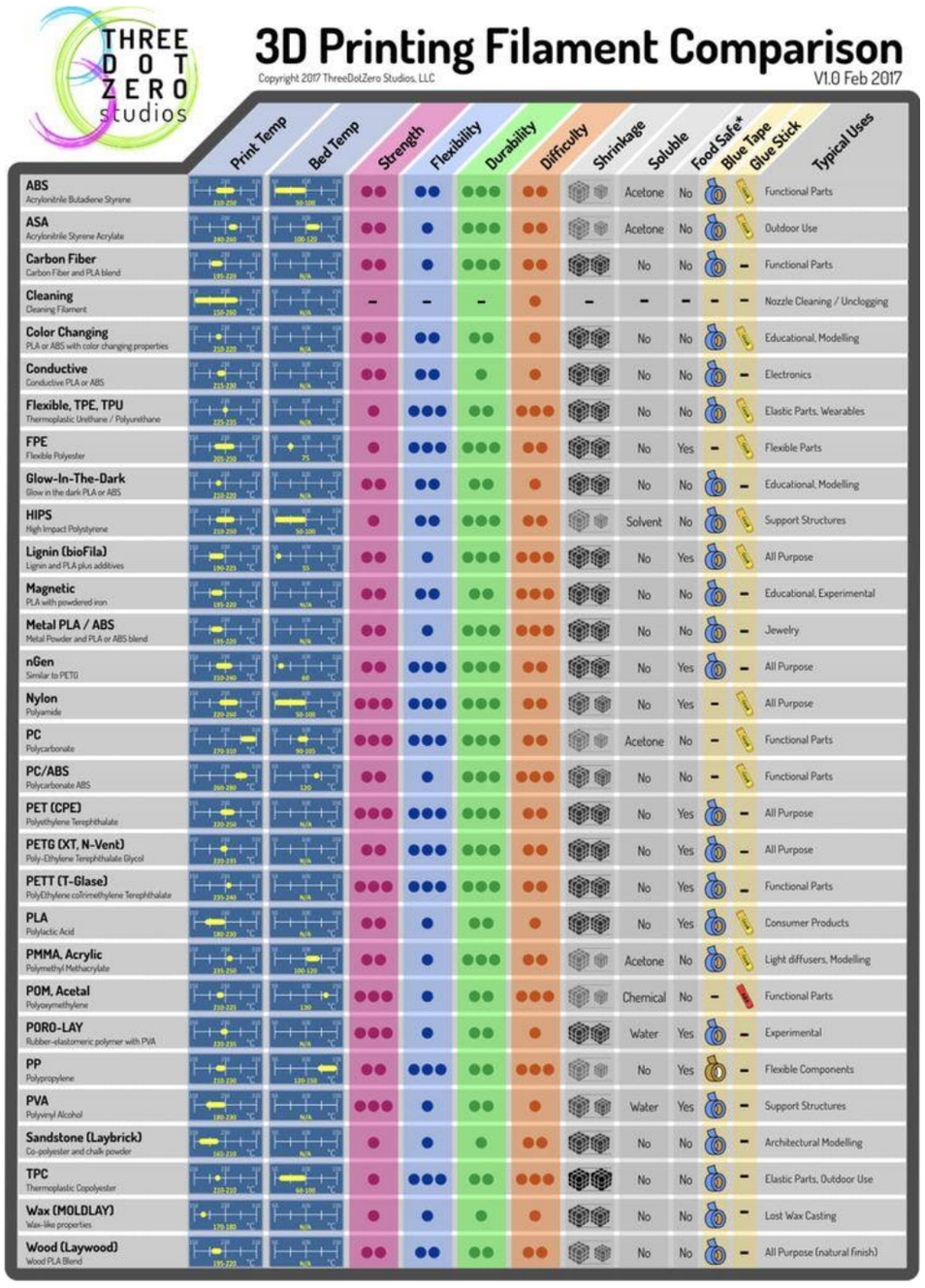


≈ 1 kg

Diámetro de **1.75 mm**

Tolerancia ± 0.03 mm

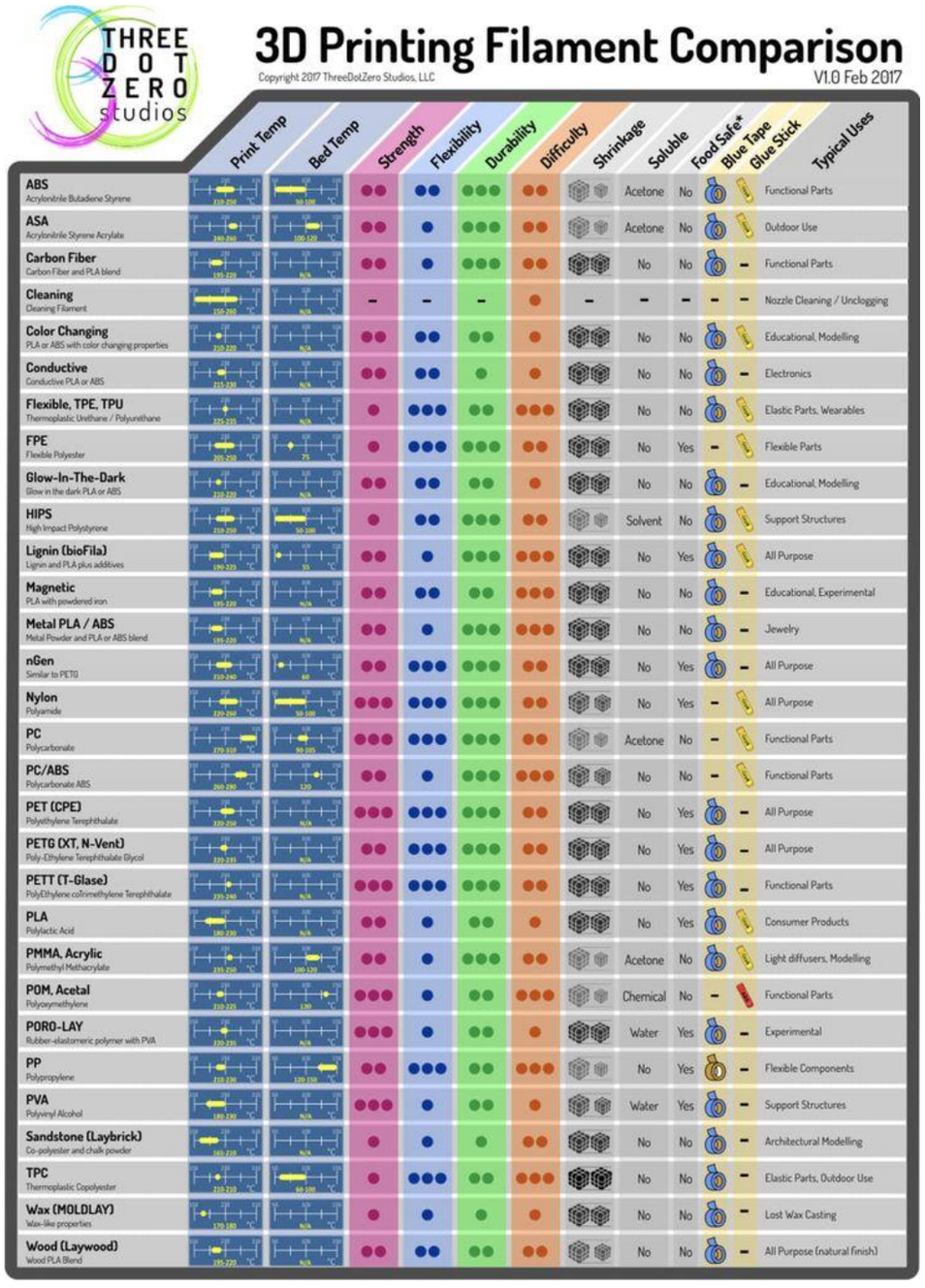
FDM – Tipos de **materiales**



	Print Temp	Bed Temp	Strength	Flexibility	Durability	Difficulty	Shrinkage	Soluble	Food Safe*	Blue Tape	Glue Stick	Typical Uses
ABS Acrylonitrile Butadiene Styrene	210-250 °C	90-110 °C	●●	●●	●●●	●●		Acetone	No		–	Functional Parts
ASA Acrylonitrile Styrene Acrylate	240-260 °C	100-120 °C	●●	●	●●●	●●		Acetone	No		–	Outdoor Use
Carbon Fiber Carbon Fiber and PLA blend	195-235 °C	N/A	●●	●	●●●	●●		No	No		–	Functional Parts
Cleaning Cleaning Filament	150-260 °C	N/A	–	–	–	●	–	–	–	–	–	Nozzle Cleaning / Unclogging
Color Changing PLA or ABS with color changing properties	210-220 °C	N/A	●●	●●	●●	●		No	No		–	Educational, Modelling
Conductive Conductive PLA or ABS	215-230 °C	N/A	●●	●●	●	●		No	No		–	Electronics
Flexible, TPE, TPU Thermoplastic Urethane / Polyurethane	220-235 °C	N/A	●	●●●	●●	●●●		No	No		–	Elastic Parts, Wearables
FPE Flexible Polyester	205-250 °C	75 °C	●	●●●	●●●	●●		No	Yes	–	–	Flexible Parts
Glow-In-The-Dark Glow in the dark PLA or ABS	210-220 °C	N/A	●●	●●	●●	●		No	No		–	Educational, Modelling
HIPS High Impact Polystyrene	210-250 °C	50-100 °C	●	●●	●●●	●●		Solvent	No		–	Support Structures
Lignin (bioFila) Lignin and PLA plus additives	190-225 °C	55 °C	●●	●	●●●	●●●		No	Yes		–	All Purpose
Magnetic PLA with powdered iron	195-220 °C	N/A	●●	●●	●●	●●●		No	No		–	Educational, Experimental
Metal PLA / ABS Metal Powder and PLA or ABS blend	195-220 °C	N/A	●●	●	●●●	●●●		No	No		–	Jewelry
nGen Similar to PETG	210-240 °C	60 °C	●●	●●●	●●●	●●		No	Yes		–	All Purpose
Nylon Polyamide	220-260 °C	50-100 °C	●●●	●●●	●●●	●●		No	Yes	–	–	All Purpose
PC Polycarbonate	270-310 °C	90-105 °C	●●●	●●●	●●●	●●		Acetone	No	–	–	Functional Parts
PC/ABS Polycarbonate ABS	260-280 °C	120 °C	●●	●	●●●	●●●		No	No	–	–	Functional Parts
PET (CPE) Polyethylene Terephthalate	220-250 °C	N/A	●●●	●●●	●●●	●●		No	Yes		–	All Purpose
PETG (XT, N-Vent) Poly-Ethylene Terephthalate Glycol	220-235 °C	N/A	●●	●●●	●●●	●●		No	Yes		–	All Purpose
PETT (T-Glase) PolyEthylene coTrimethylene Terephthalate	235-240 °C	N/A	●●●	●●●	●●●	●●		No	Yes		–	Functional Parts
PLA Polylactic Acid	180-230 °C	N/A	●●	●	●●	●		No	Yes		–	Consumer Products
PMMA, Acrylic Polymethyl Methacrylate	235-250 °C	100-120 °C	●●	●	●●●	●●		Acetone	No		–	Light diffusers, Modelling
POM, Acetal Polyoxymethylene	210-225 °C	130 °C	●●●	●	●●	●●●		Chemical	No	–	–	Functional Parts
PORO-LAY Rubber-elastomeric polymer with PVA	210-235 °C	N/A	●●●	●	●●	●		Water	Yes		–	Experimental
PP Polypropylene	210-230 °C	120-150 °C	●●	●●●	●●	●●●		No	Yes		–	Flexible Components
PVA Polyvinyl Alcohol	180-230 °C	N/A	●●●	●	●●	●		Water	Yes		–	Support Structures
Sandstone (Laybrick) Co-polyester and chalk powder	185-210 °C	N/A	●	●	●	●●		No	No		–	Architectural Modelling
TPC Thermoplastic Copolyester	210-270 °C	60-100 °C	●	●●●	●●	●●●		No	No		–	Elastic Parts, Outdoor Use
Wax (MOLDLAY) Wax-like properties	170-180 °C	N/A	●	●	●	●		No	No		–	Lost Wax Casting
Wood (Laywood) Wood PLA Blend	195-225 °C	N/A	●●	●●	●●	●●		No	No		–	All Purpose (natural finish)

Enlace → [3D printing Filament Comparison]

FDM – Tipos de **materiales**

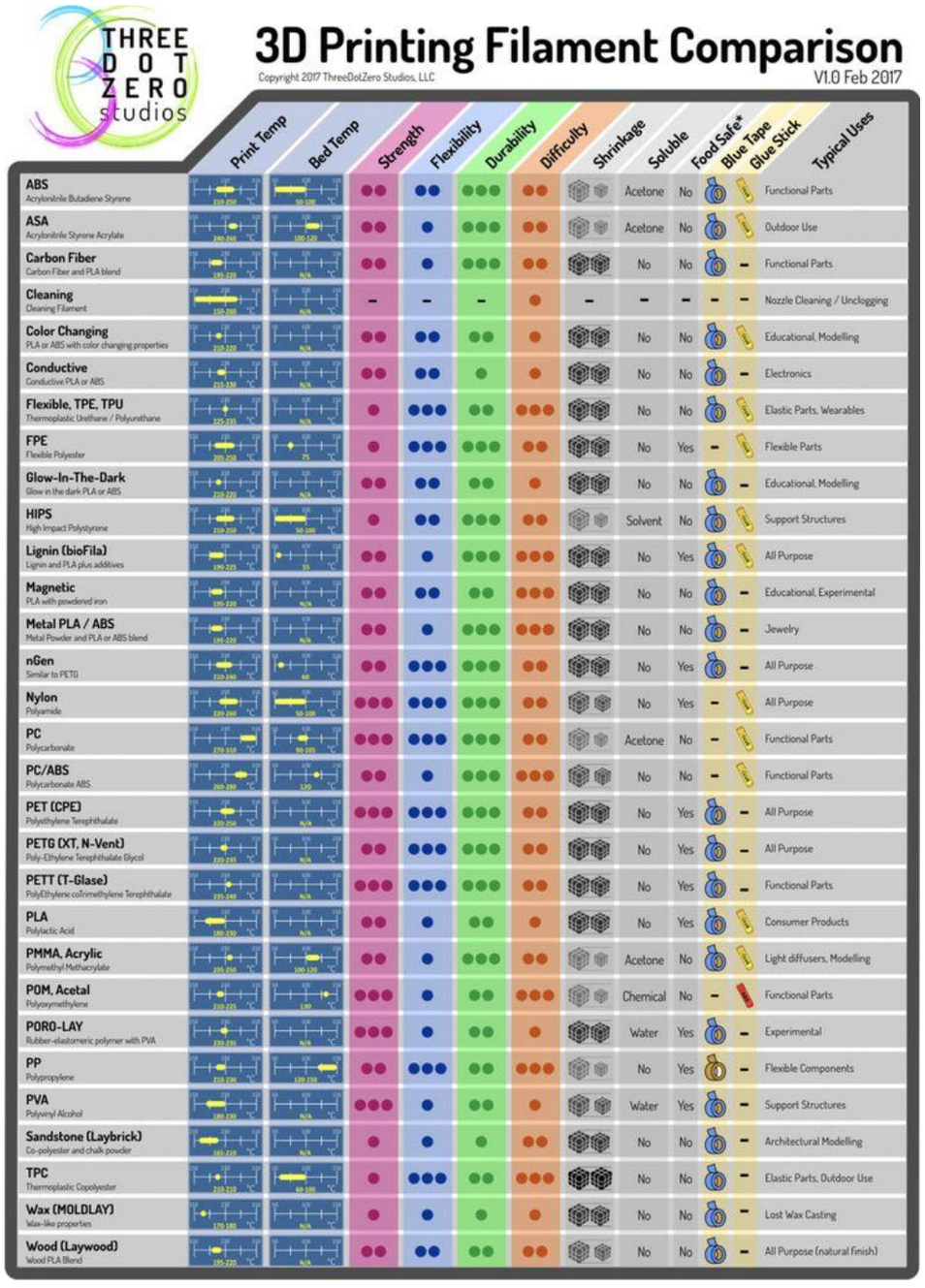


Más utilizados

- PLA
- ABS
- PETG
- ASA
- TPU
- PVA

Enlace → [3D printing Filament Comparison]

FDM – Tipos de **materiales**



Más utilizados

- PLA
- ABS
- PETG
- ASA
- TPU
- PVA

Enlace → [3D printing Filament Comparison]

FDM – Tipos de **materiales**

PLA
Ácido poliláctico

- Facilidad ✓✓
- Biodegradable
- Inodoro
- Resistencia al calor y químicos
- Duro pero frágil

PETG
Tereftalato de polietileno glicolizado

- Facilidad ✓
- Apto para alimentación*
- Alta transparencia
- Buena tenacidad

TPU
Poliuretano termoplástico

- Facilidad ✗
- Flexible
- Resistente a impactos

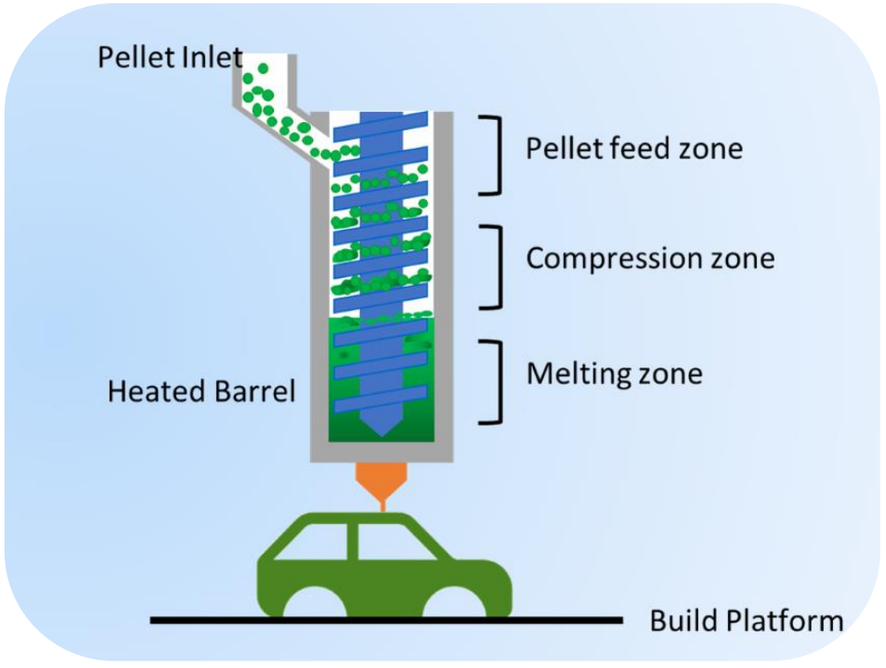
ABS
Acrilonitrilo butadieno estireno

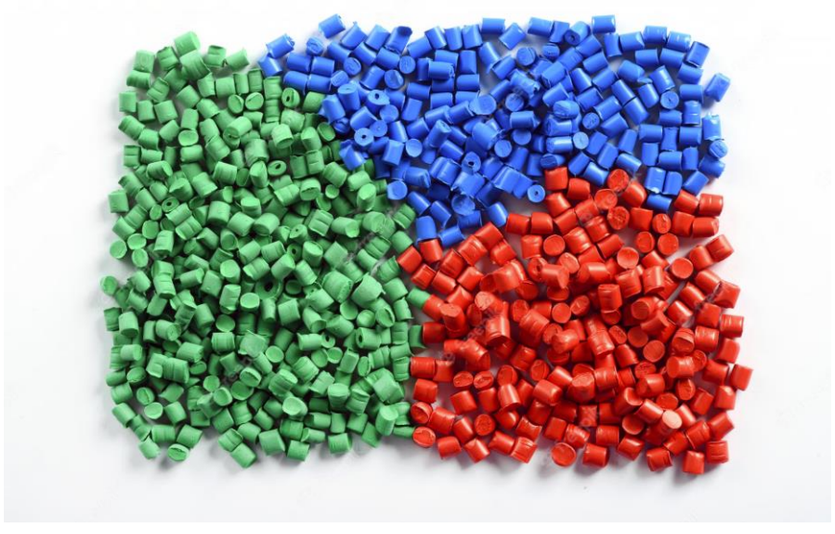
- Facilidad ✗✗
- Necesita ventilación
- Alta tenacidad

FDM – **Material** de impresión

EXTRUSOR DE PELLETS







Reutilización de material

Más materiales disponibles

Capacidad de material alta

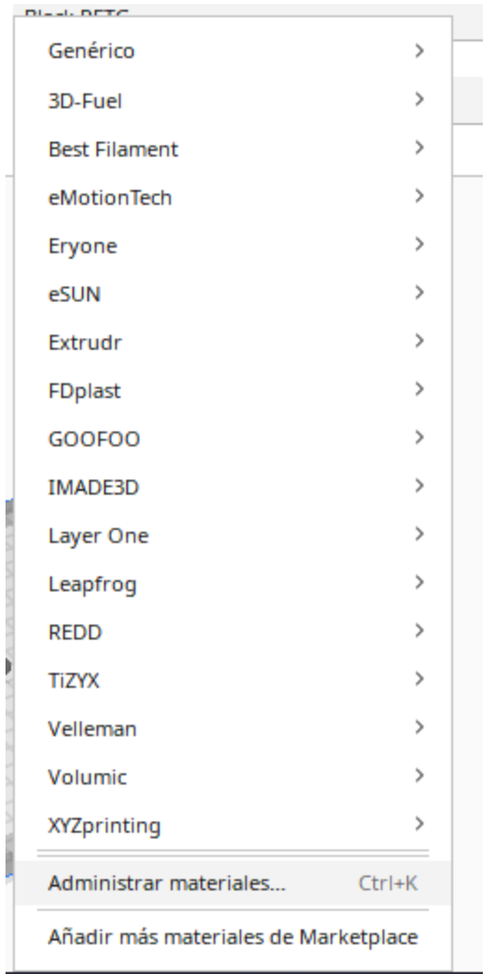
FDM – **Material** de impresión

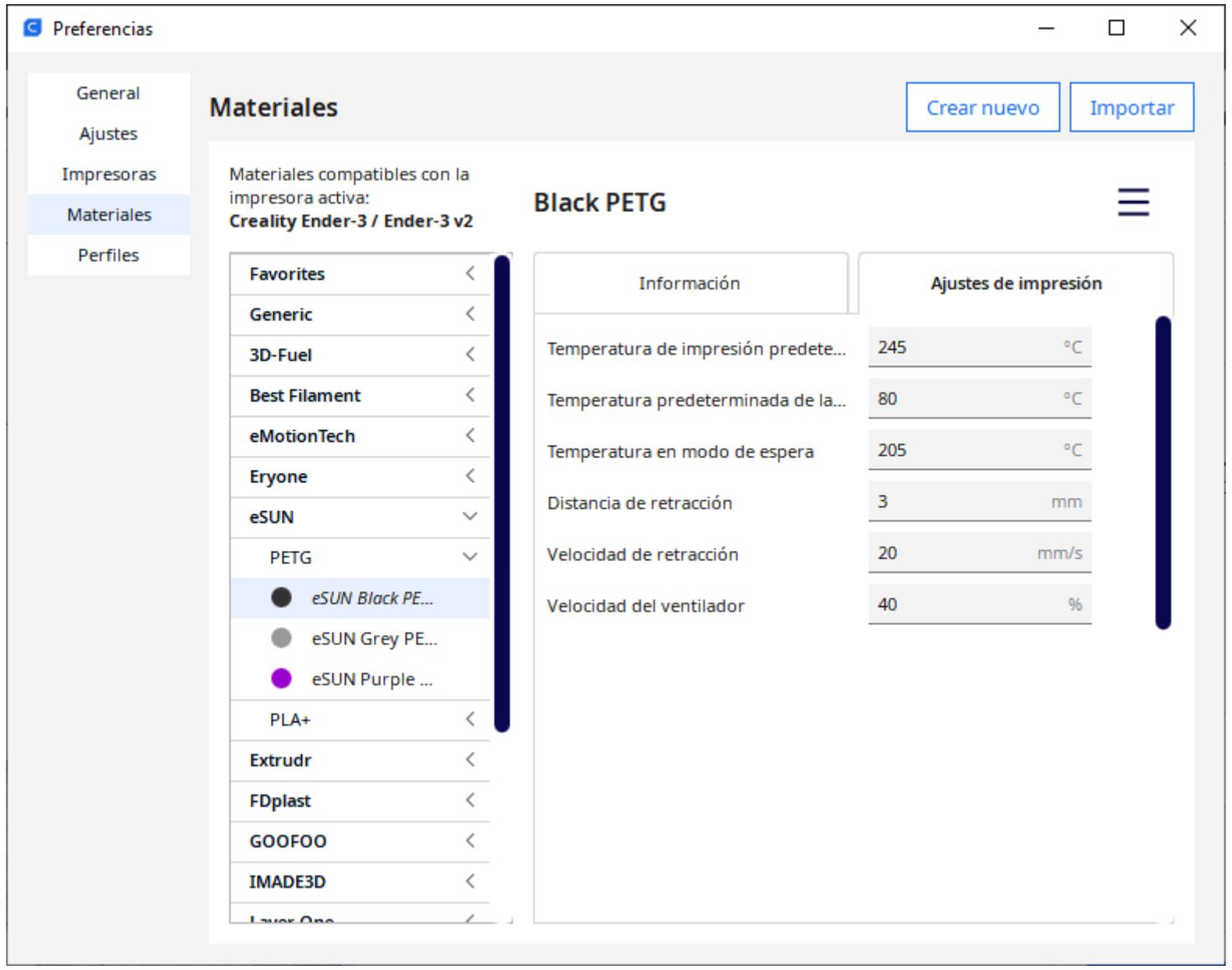


https://www.youtube.com/@DrDFlo

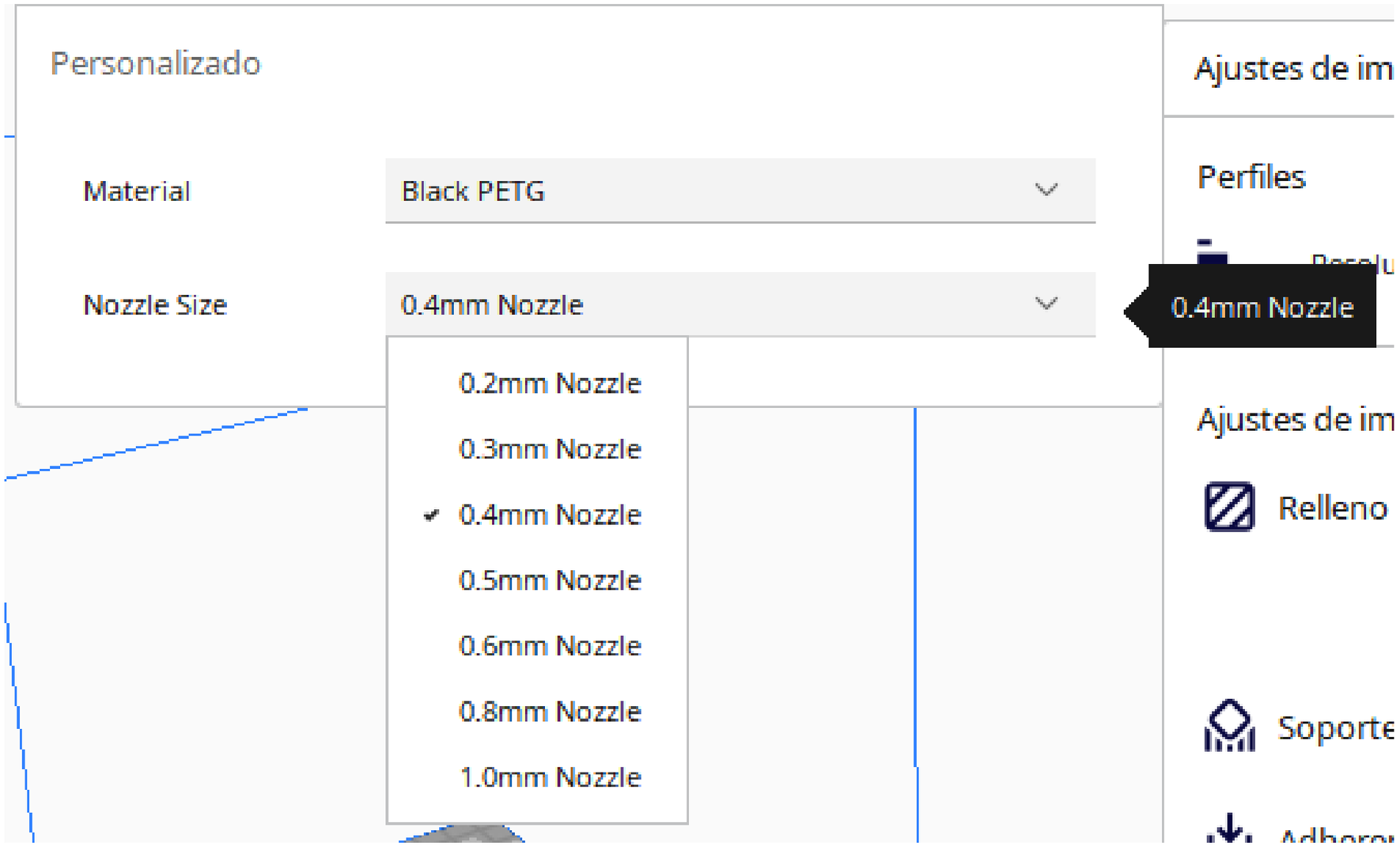
CURA – **Entorno** de trabajo

Selección material y boquilla

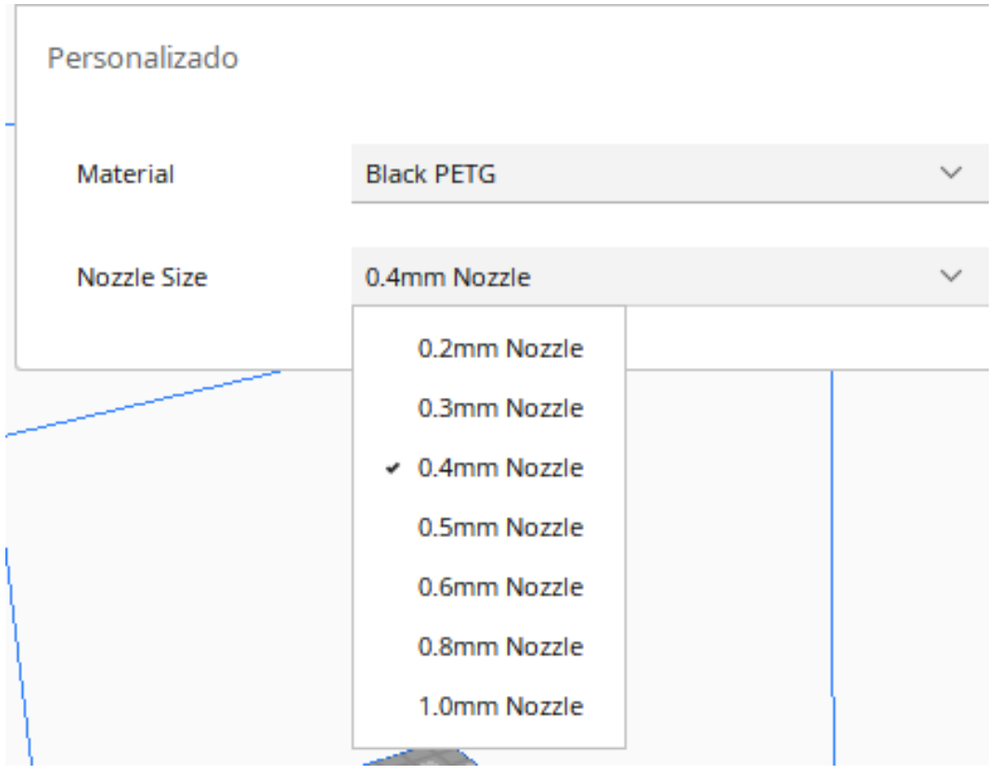


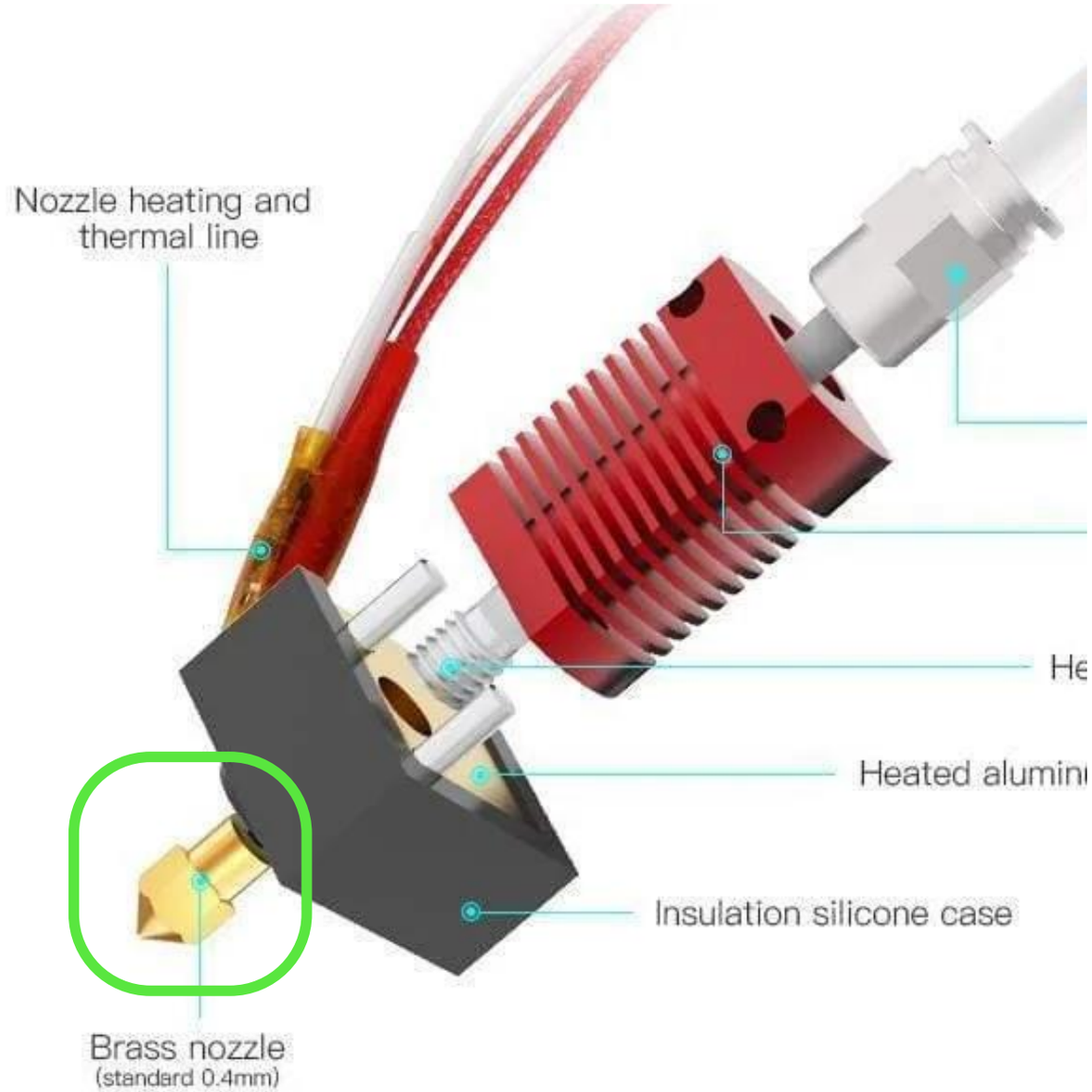


FDM – Tipos de **boquillas** (noozle)

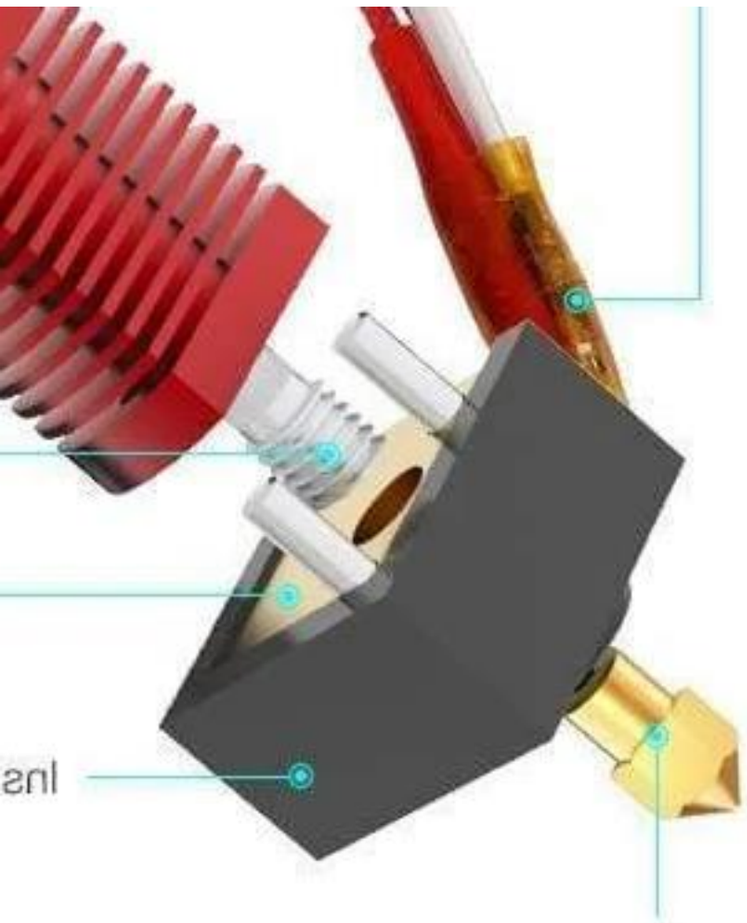


FDM – Tipos de **boquillas** (noozle)





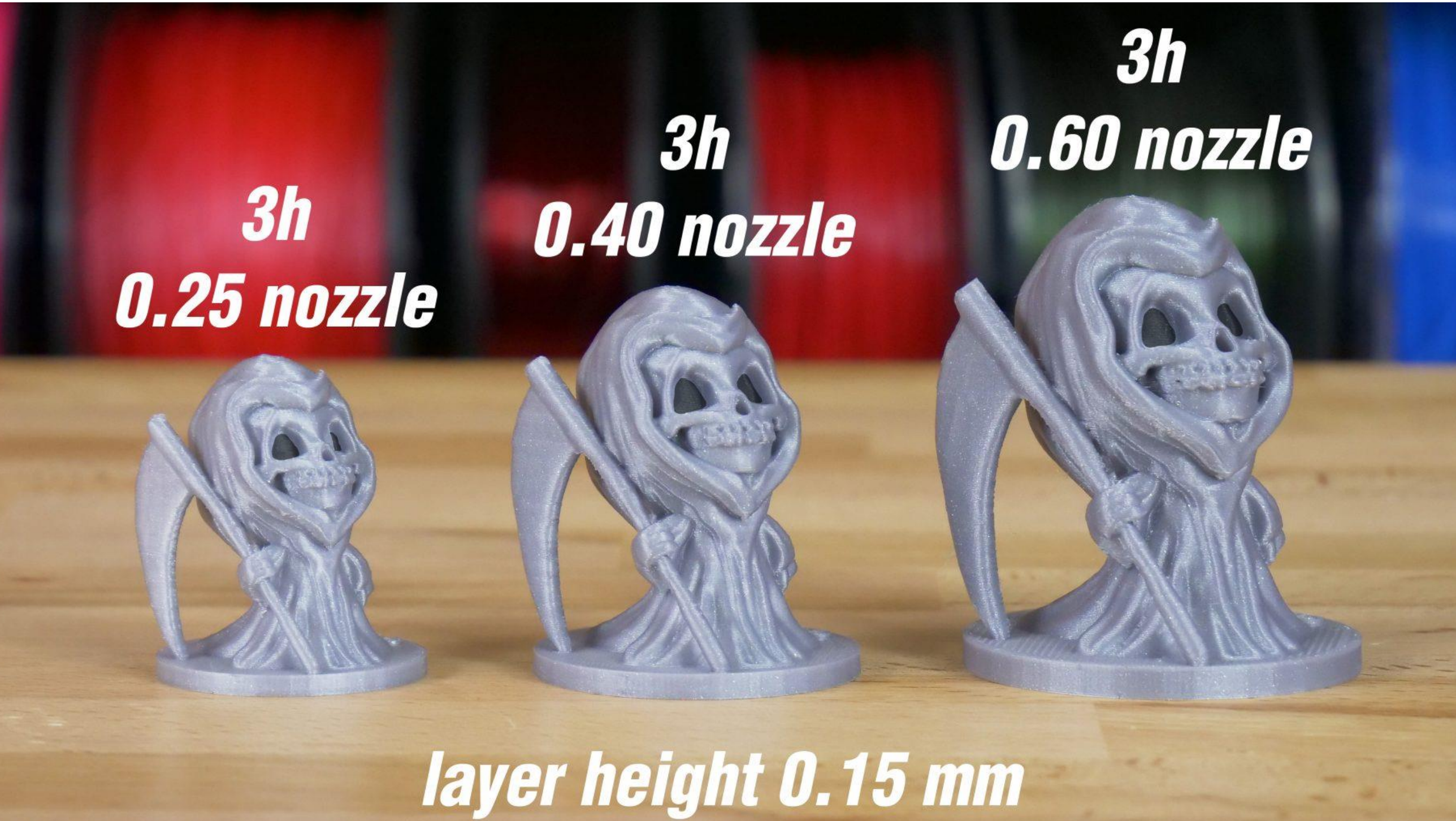
FDM – Tipos de **boquillas** (noozle)



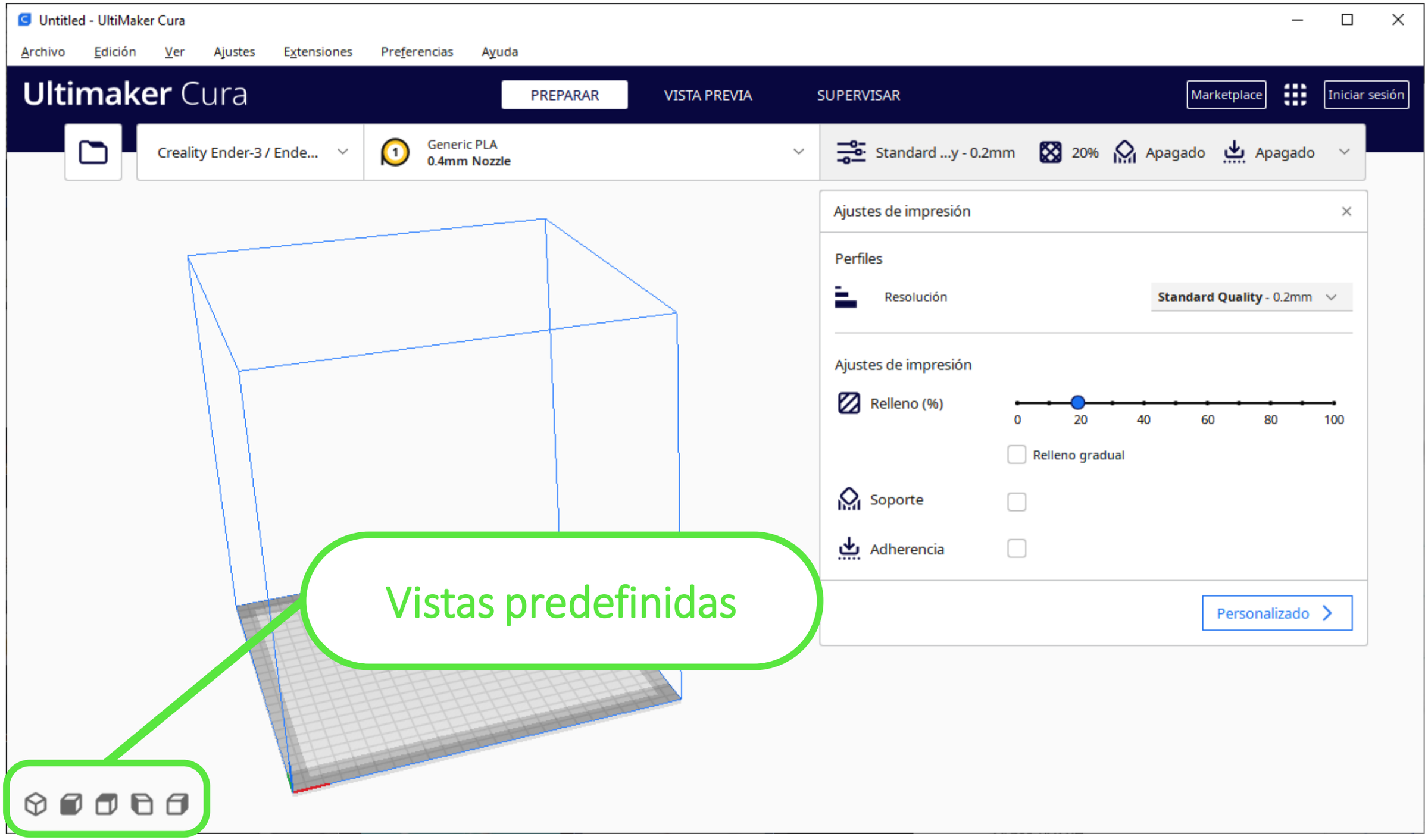




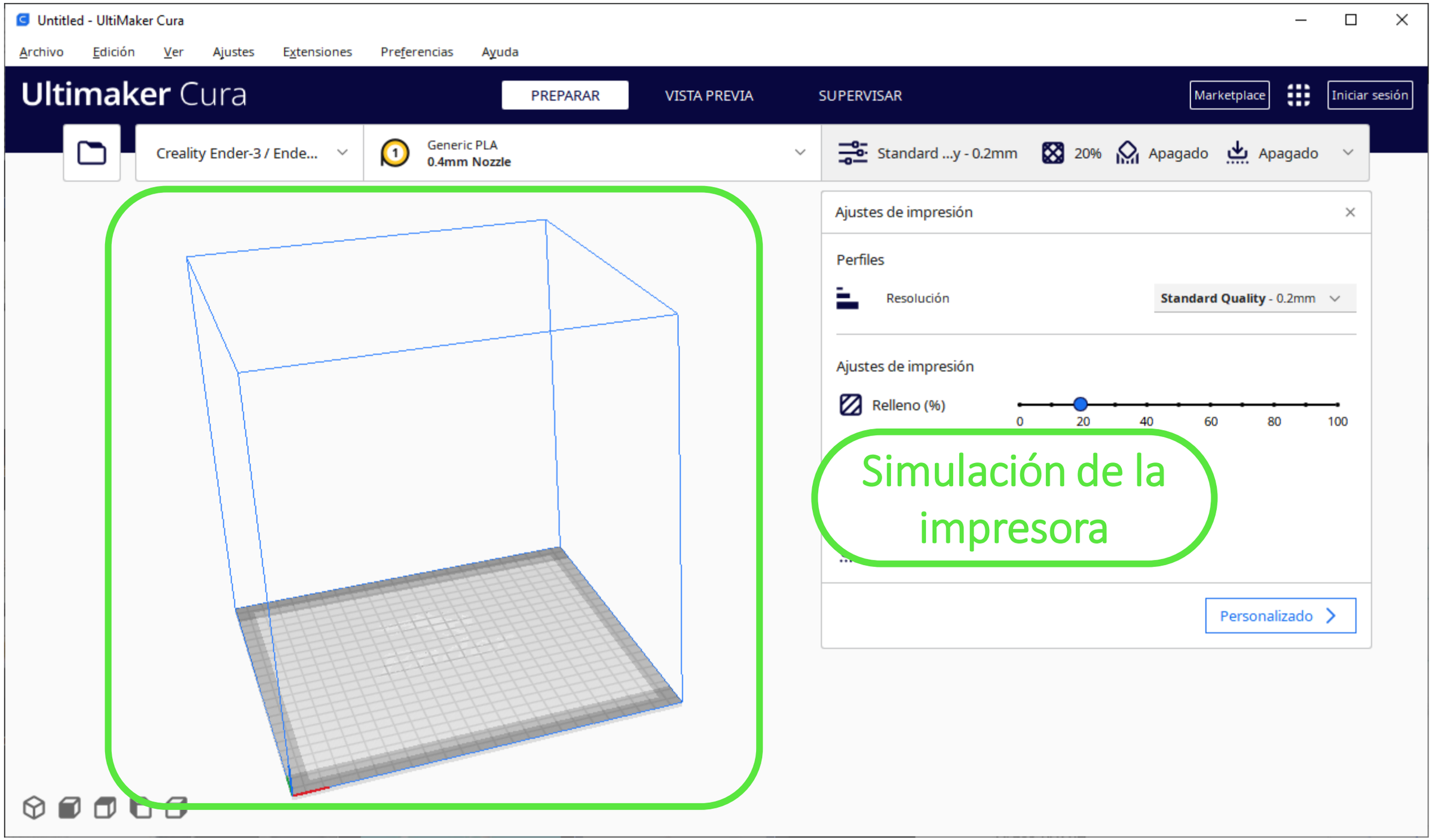
FDM – Tipos de **boquillas** (noozle)



CURA– **Entorno** de trabajo



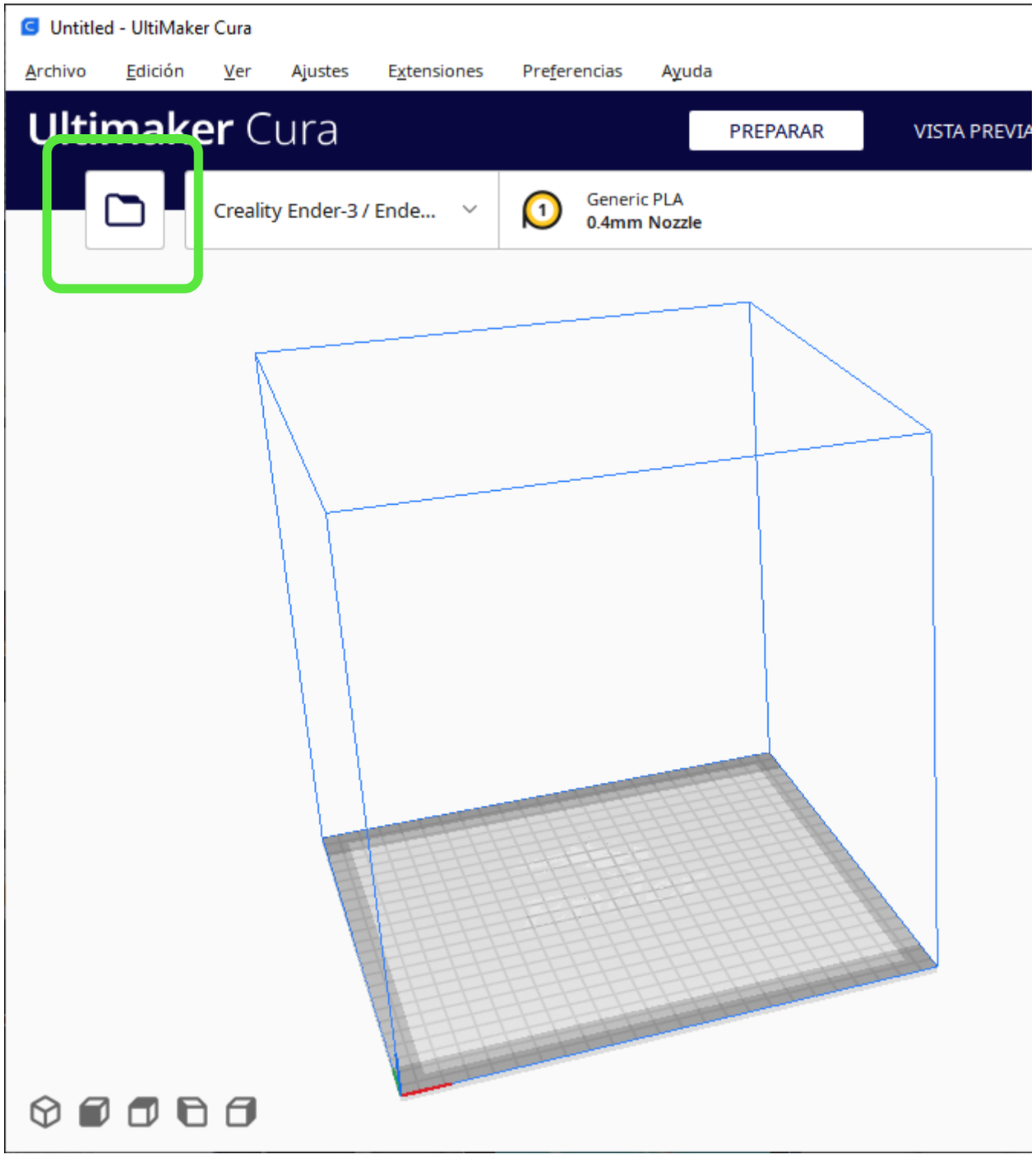
CURA– **Entorno** de trabajo



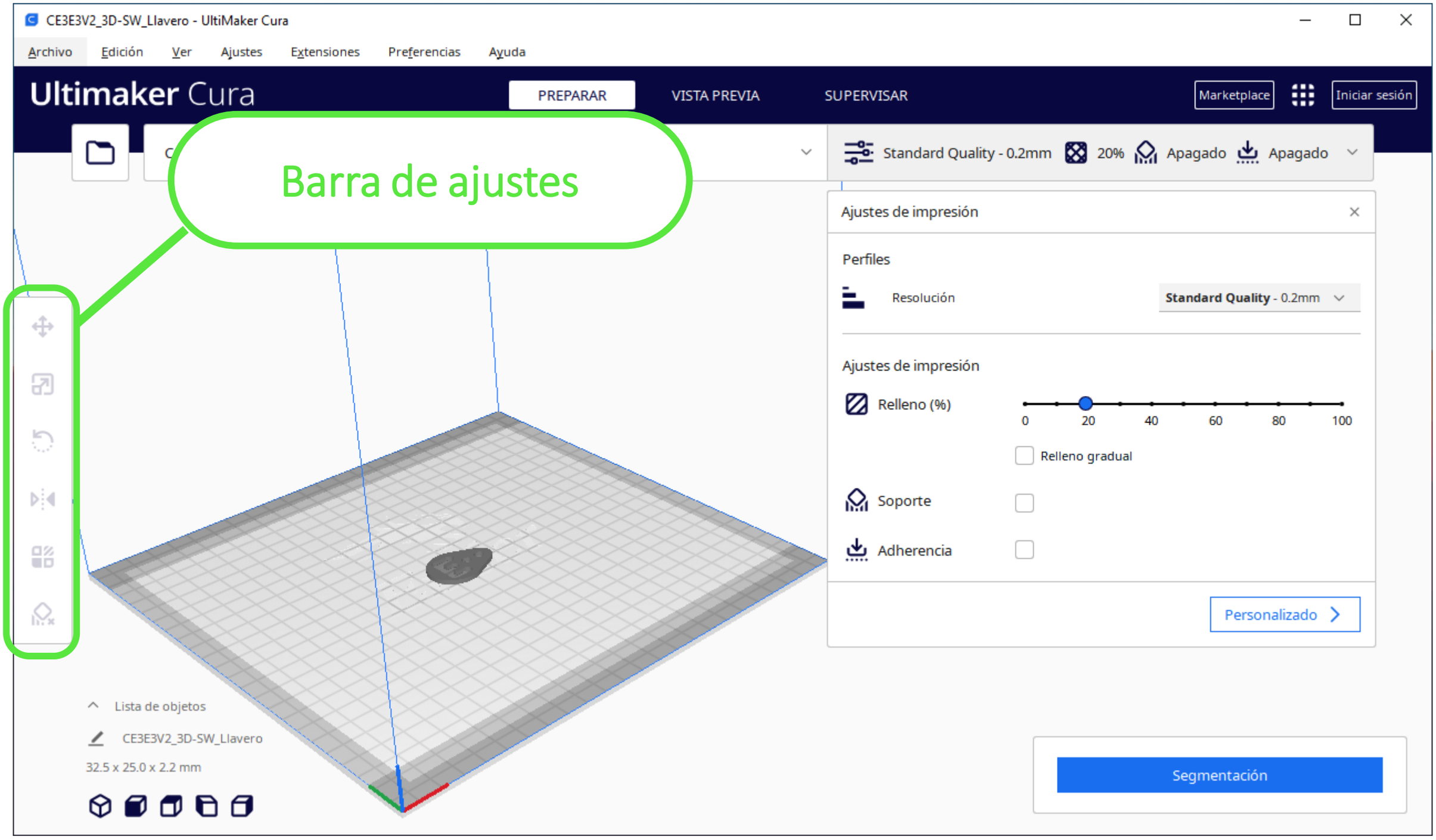
CURA– **Importar** modelo 3D



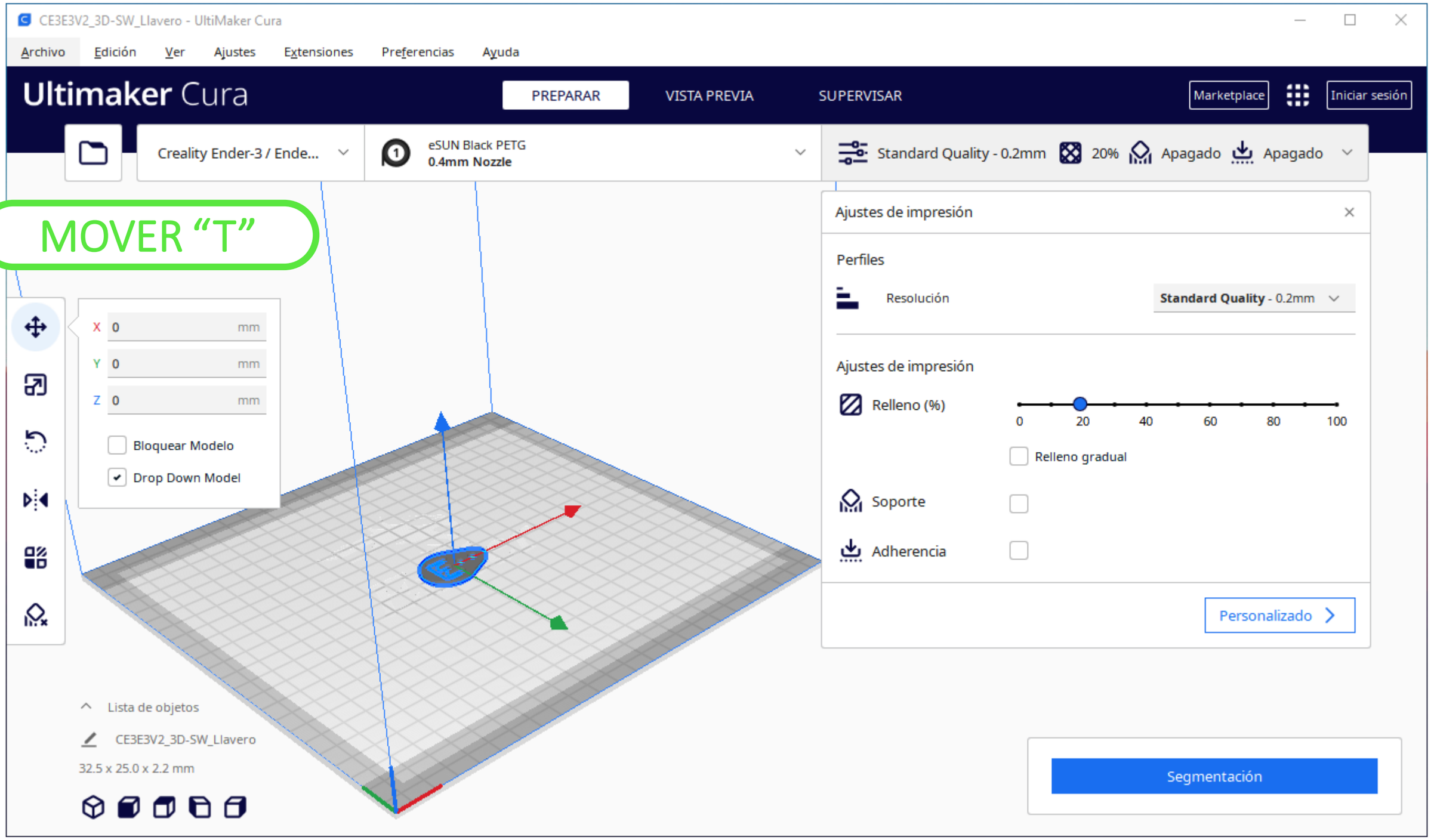
Clase 1. Ejercicio Práctico B



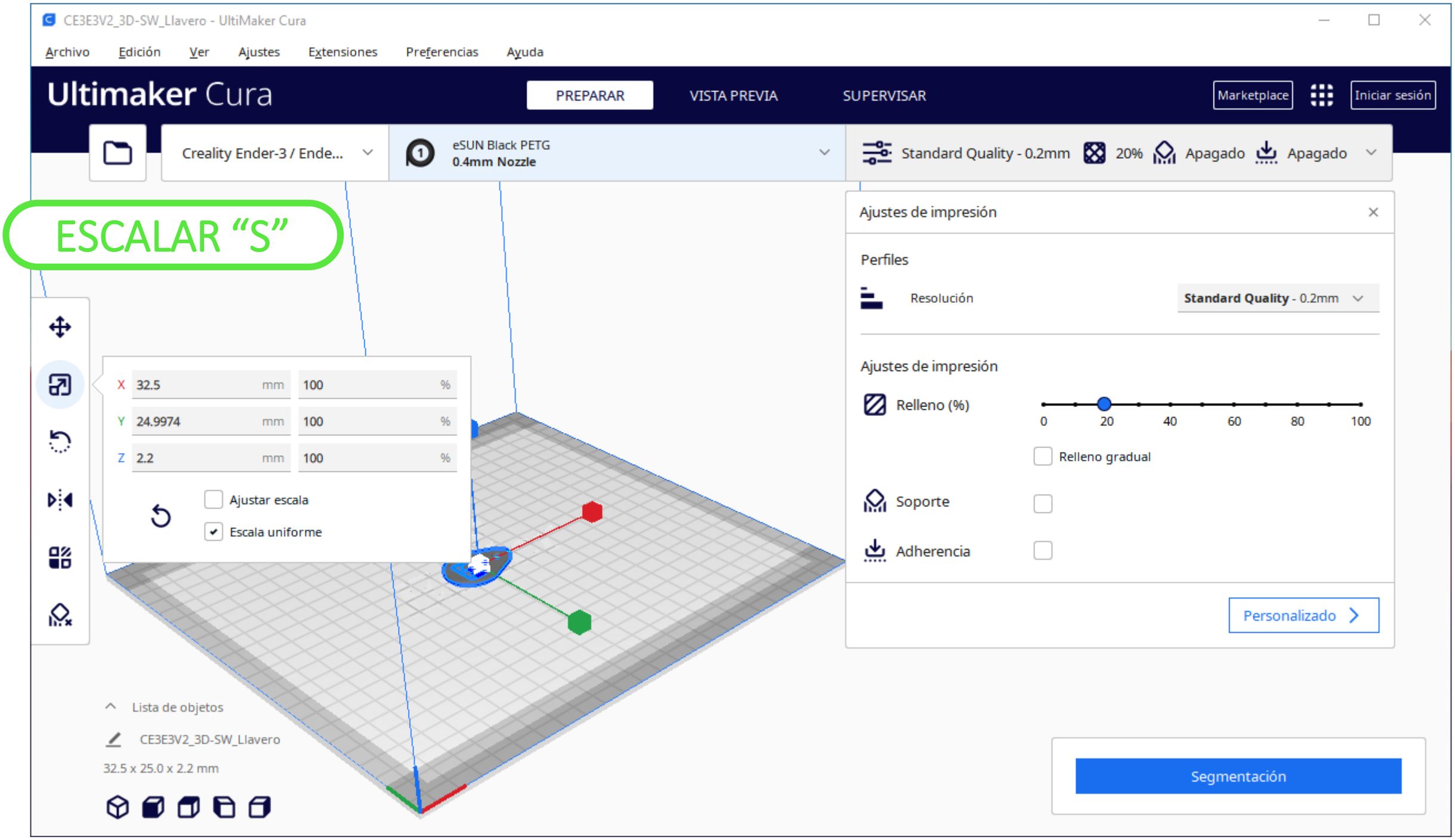
CURA– **Entorno** de trabajo



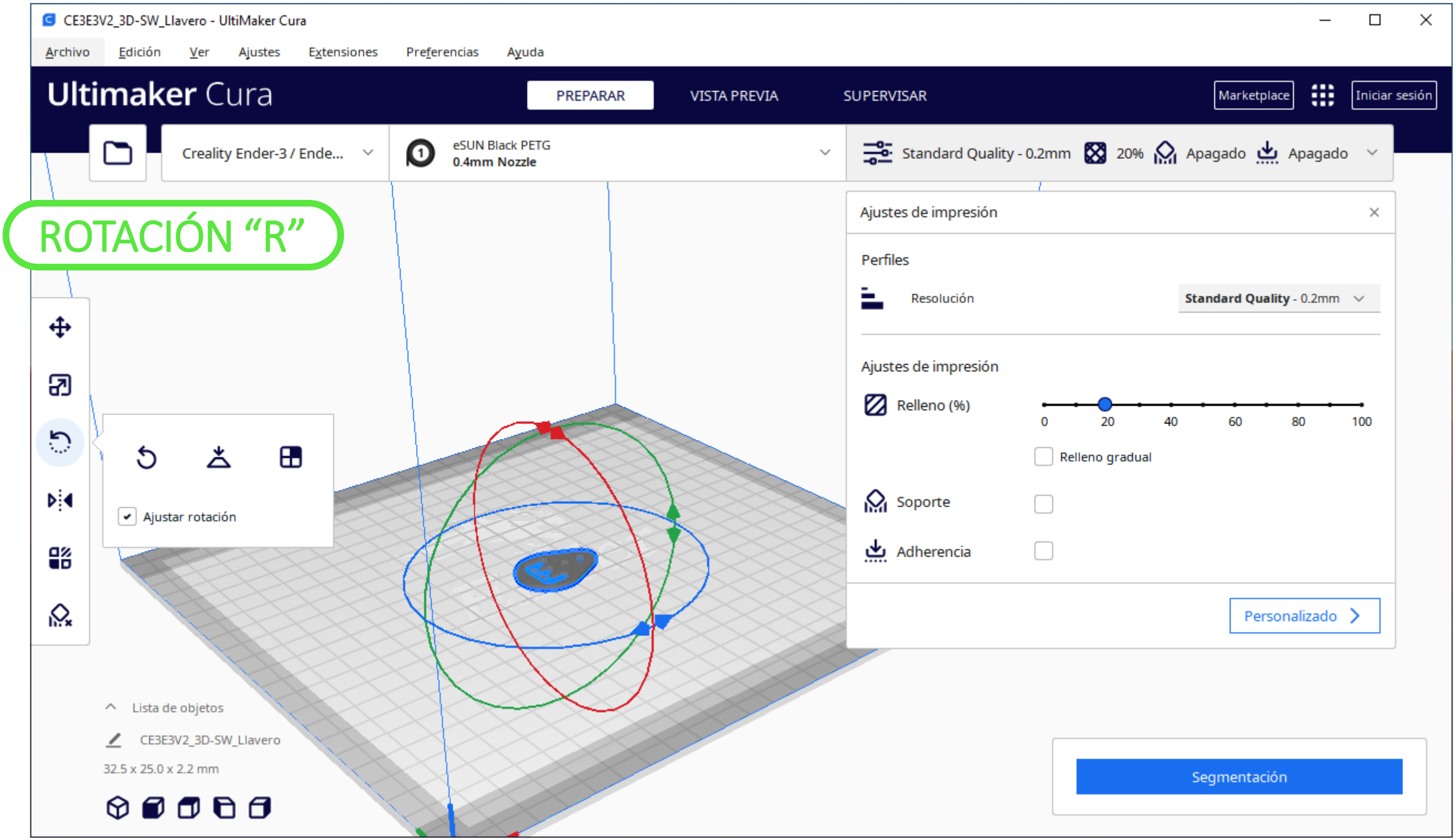
CURA– **Entorno** de trabajo



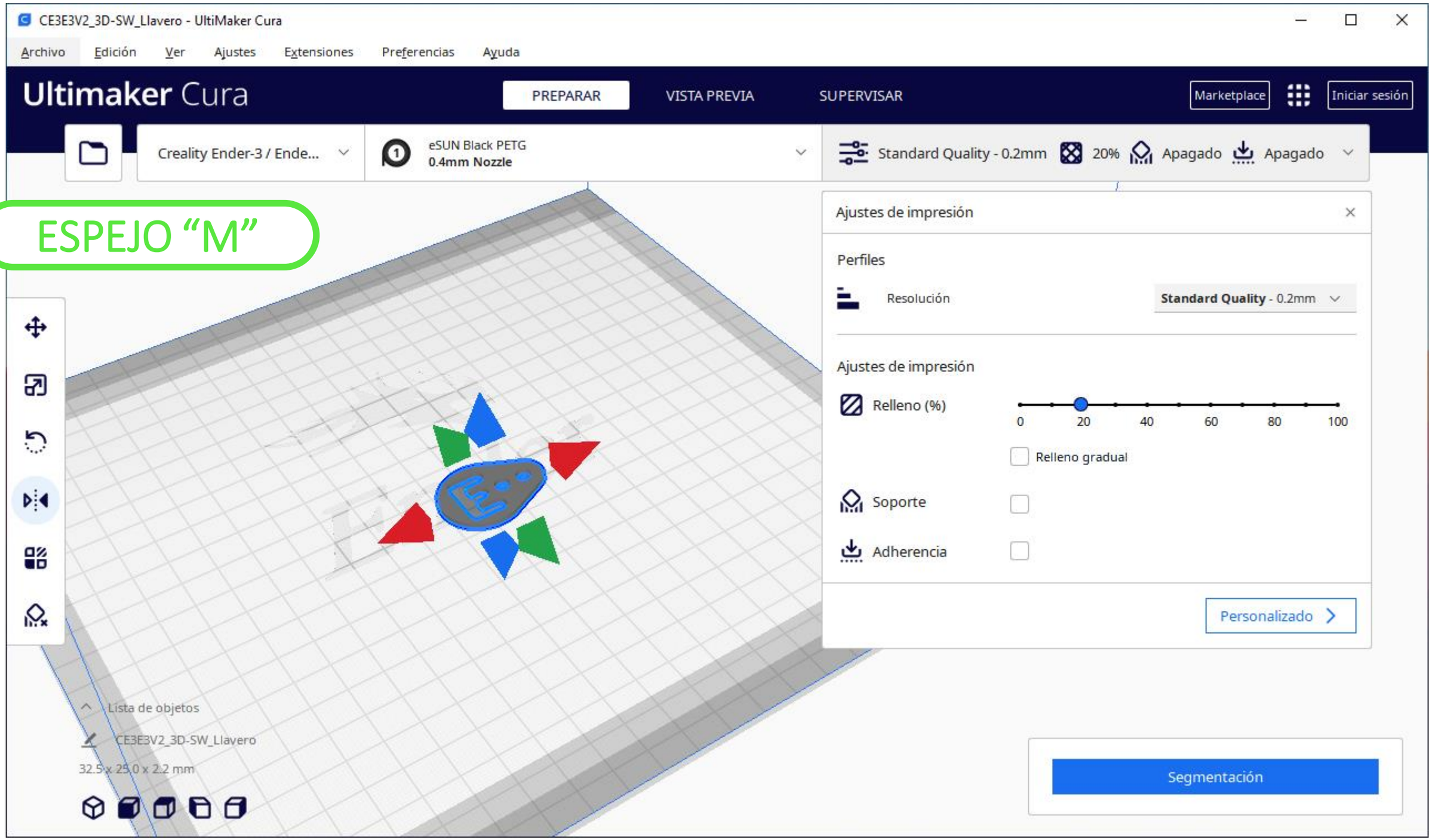
CURA– **Entorno** de trabajo



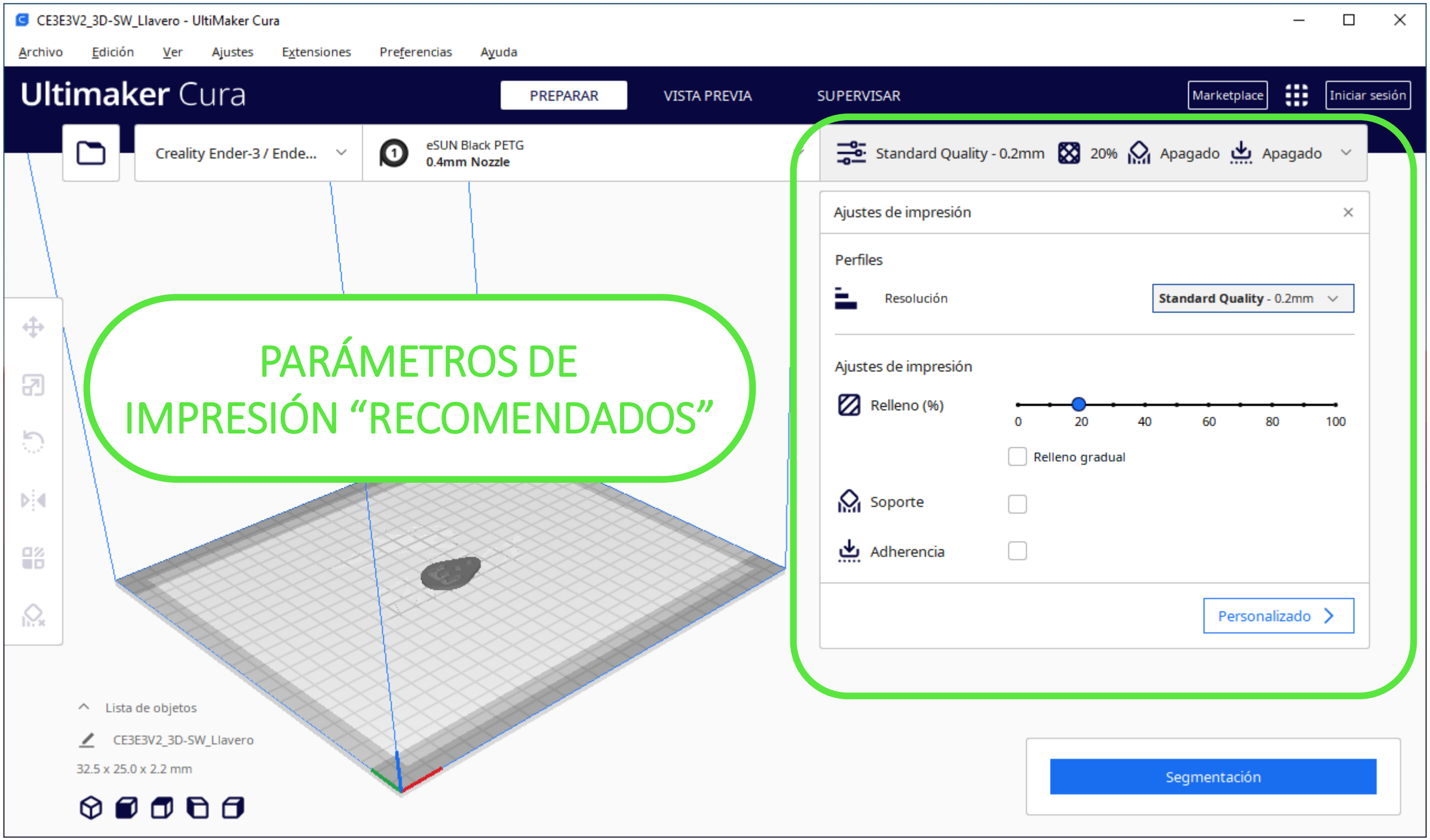
CURA– **Entorno** de trabajo



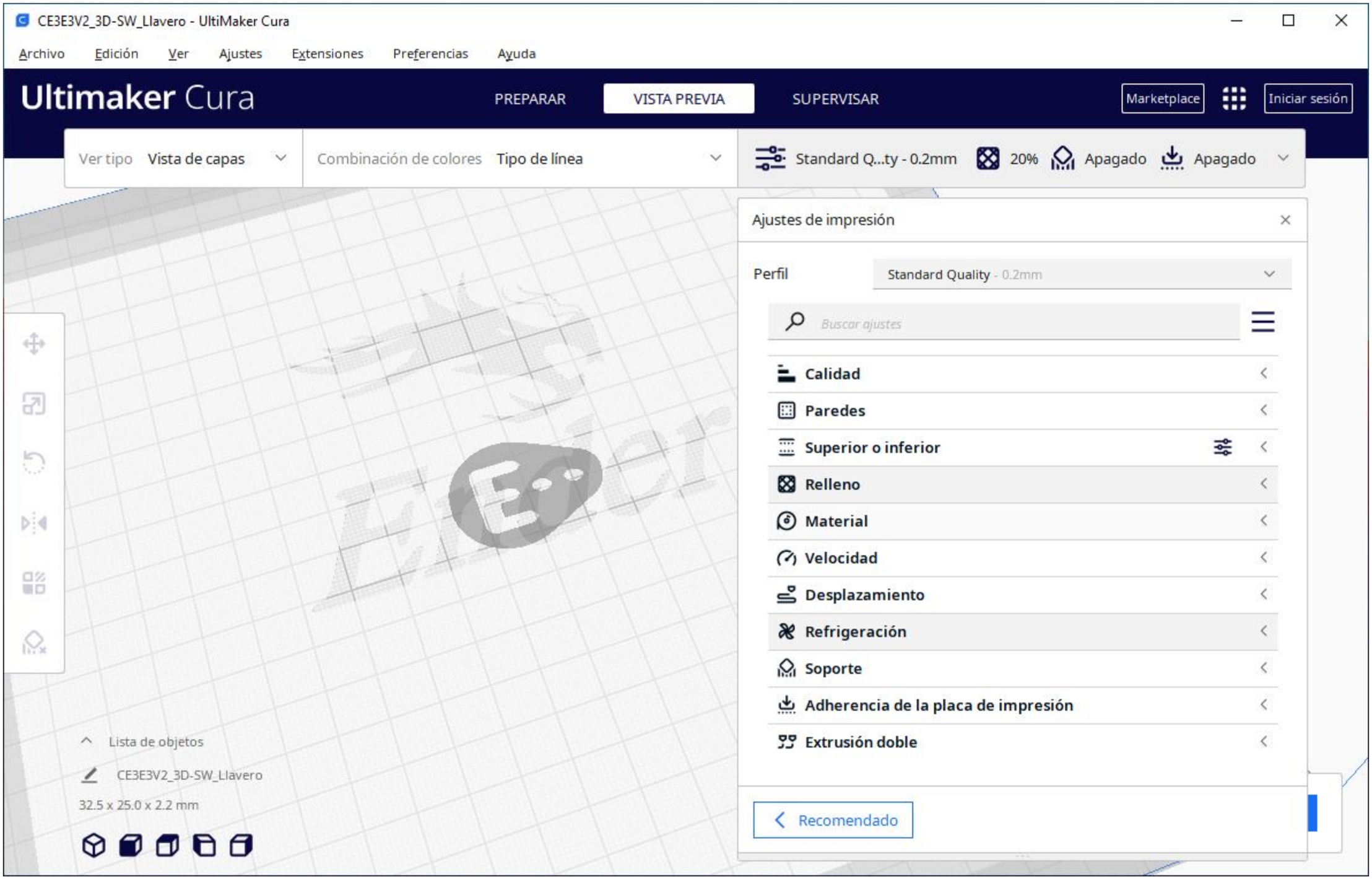
CURA– **Entorno** de trabajo



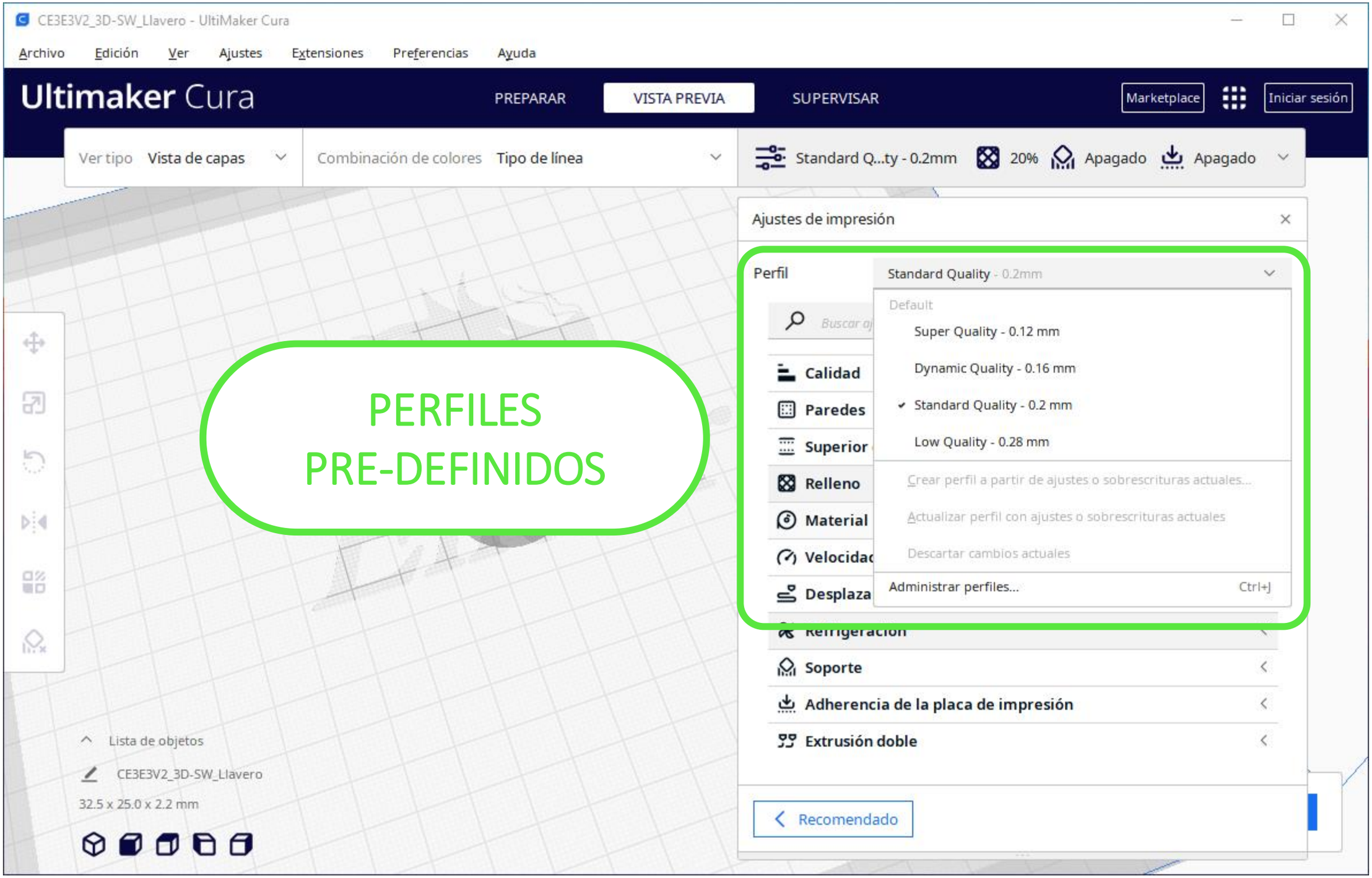
CURA– **Parámetros** de impresión



CURA– **Parámetros** de impresión



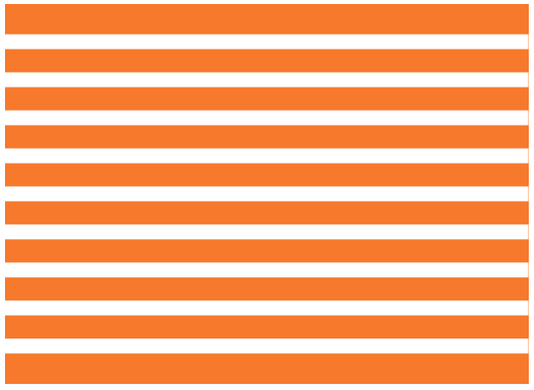
CURA– **Parámetros** de impresión



CURA– **Parámetros** de impresión

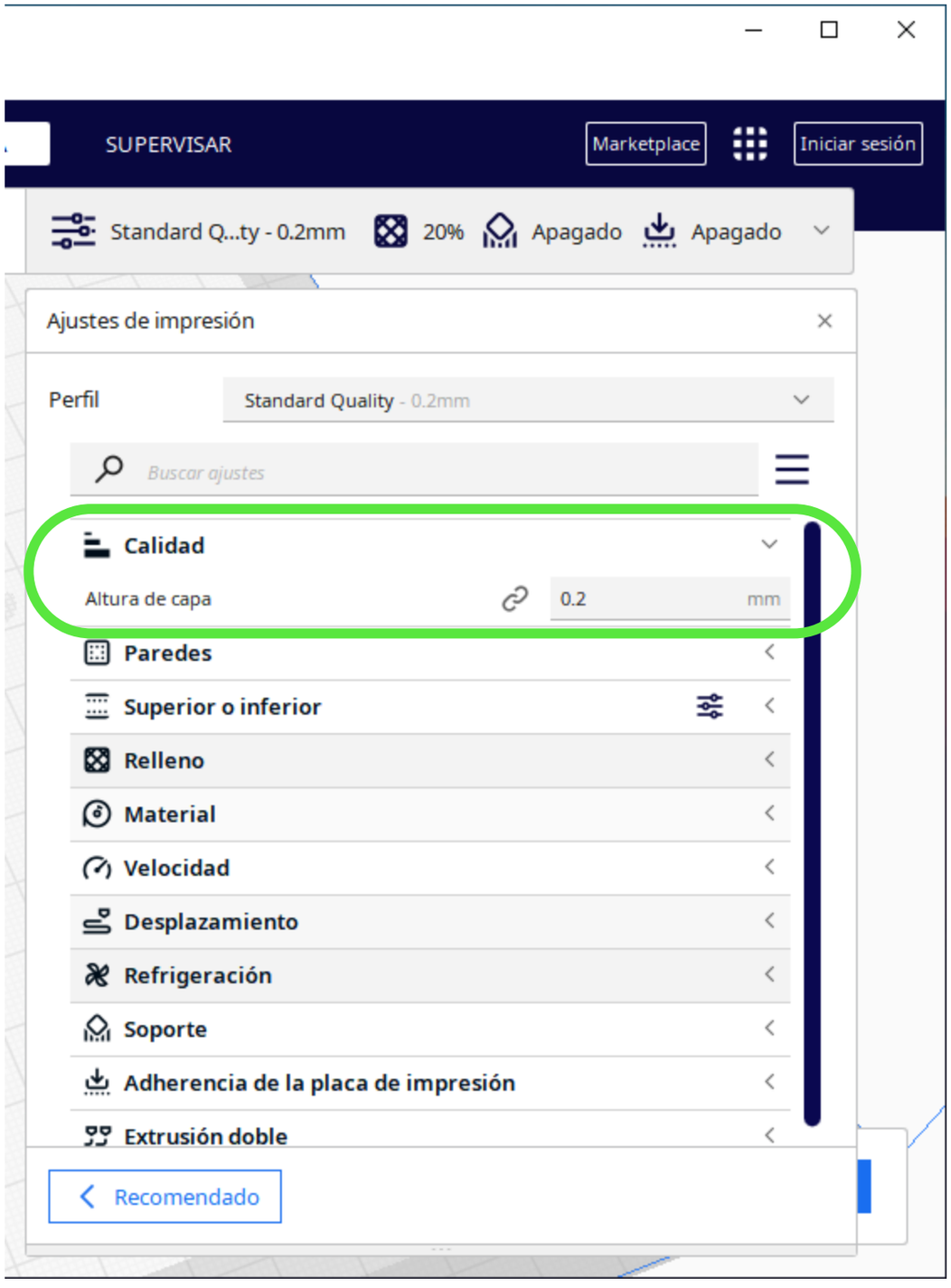
1 ALTURA DE CAPA

0.05 mm



0.3 mm

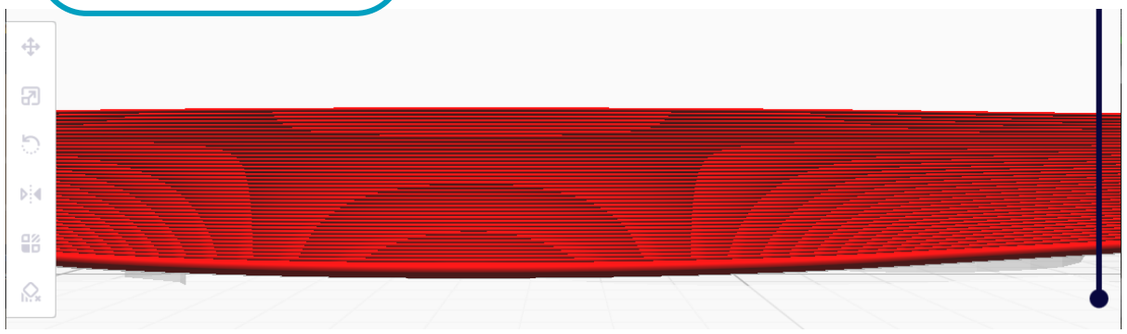




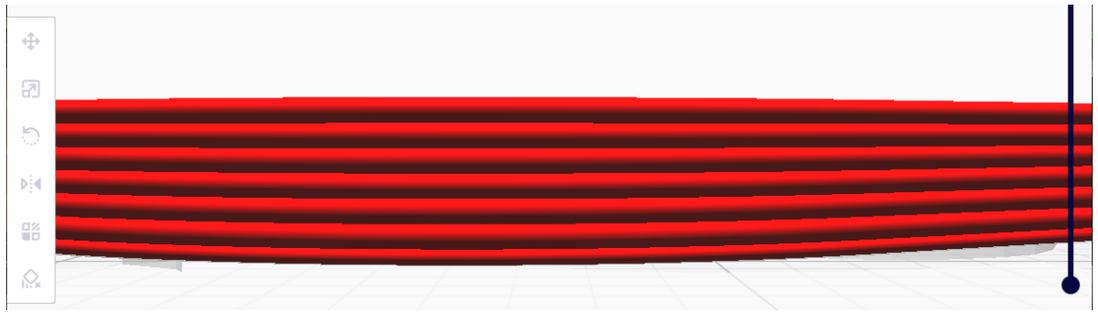
CURA– **Parámetros** de impresión

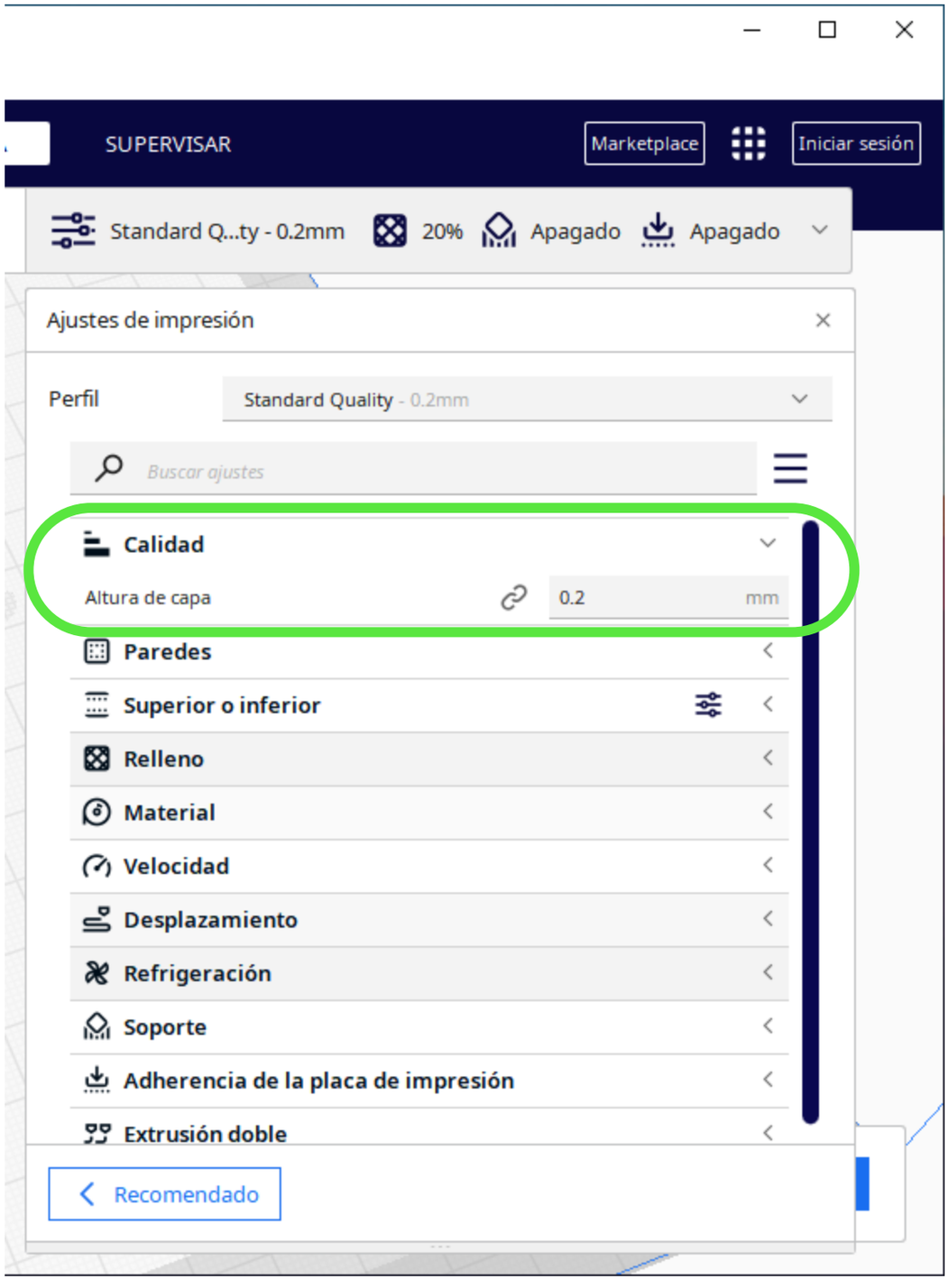
1 ALTURA DE CAPA

0.05 mm



0.3 mm

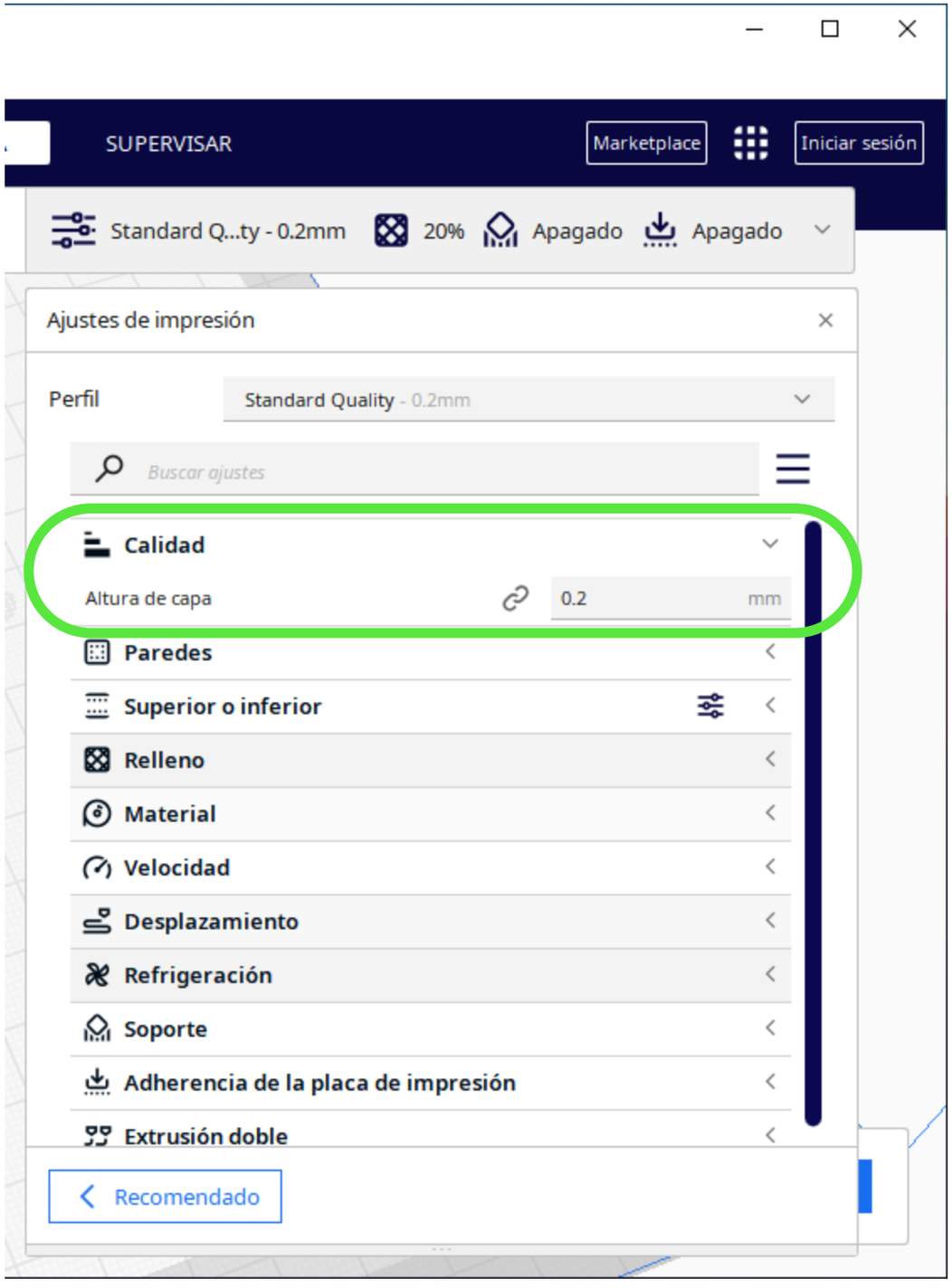




CURA– **Parámetros** de impresión

1 ALTURA DE CAPA





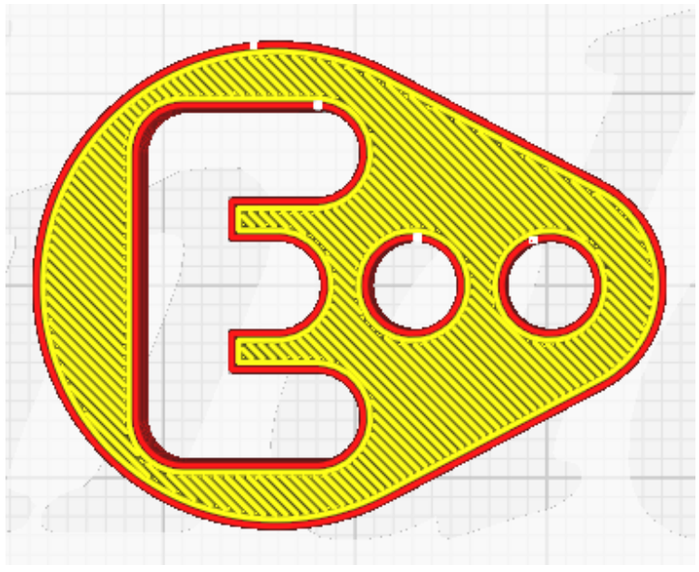
Boquilla 0,4mm

Altura de capa recomendada:
0,08 (20%) – **0,32** (80%) mm

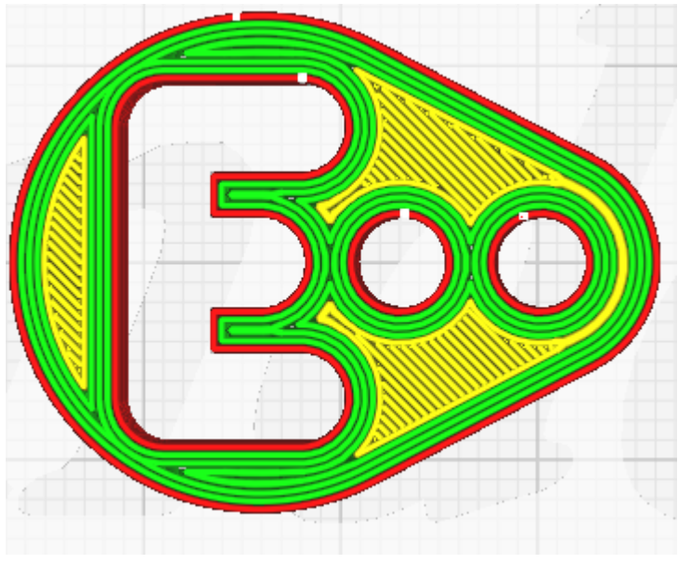
CURA– **Parámetros** de impresión

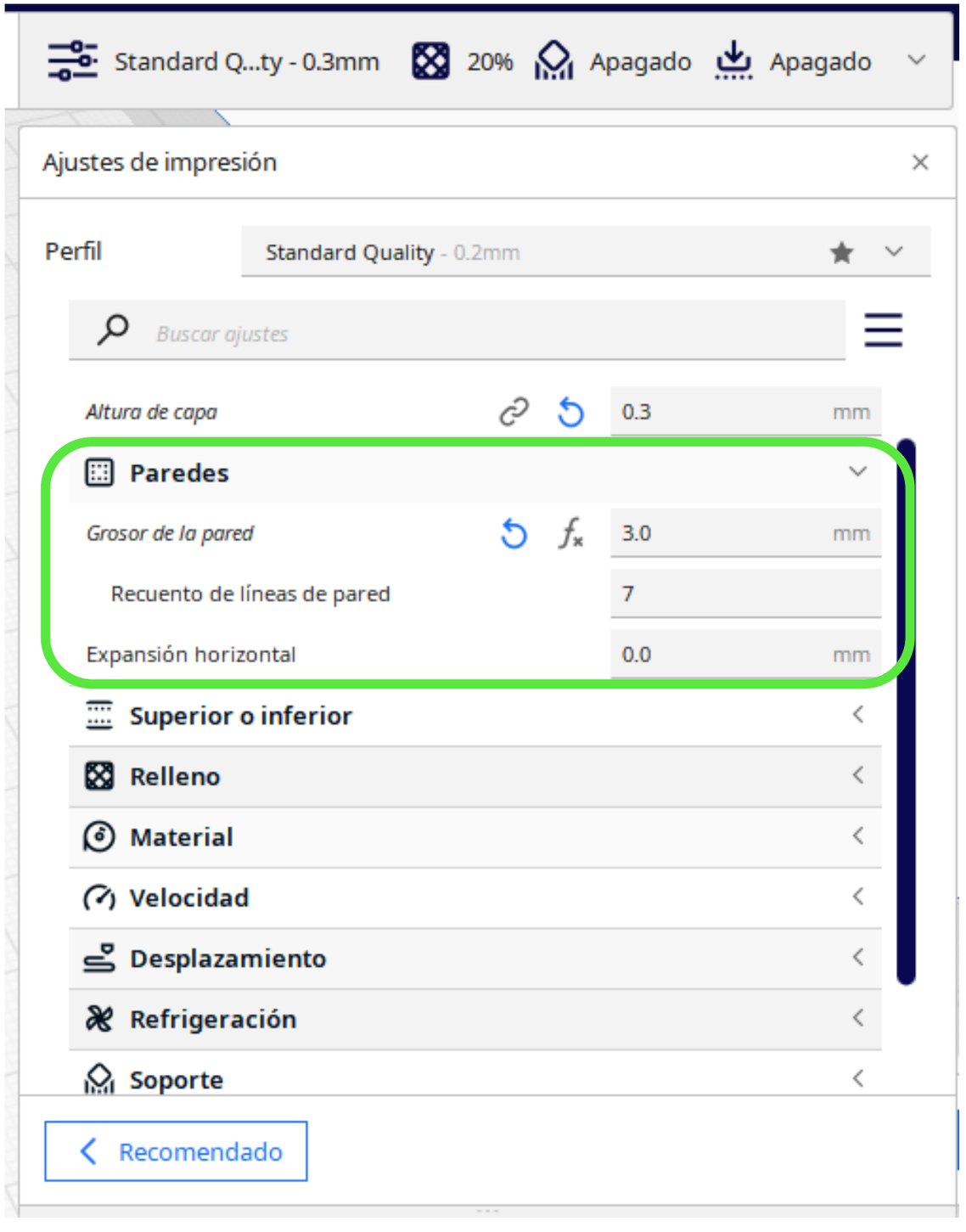
2 PAREDES

Paredes = 1



Paredes = 4

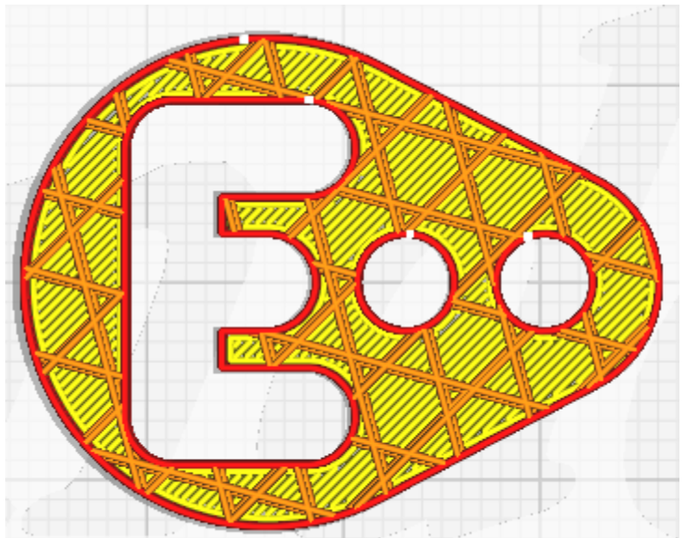




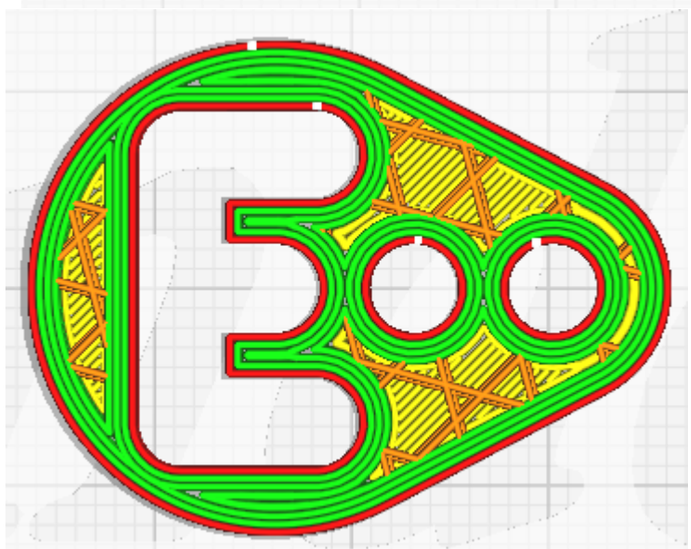
CURA– **Parámetros** de impresión

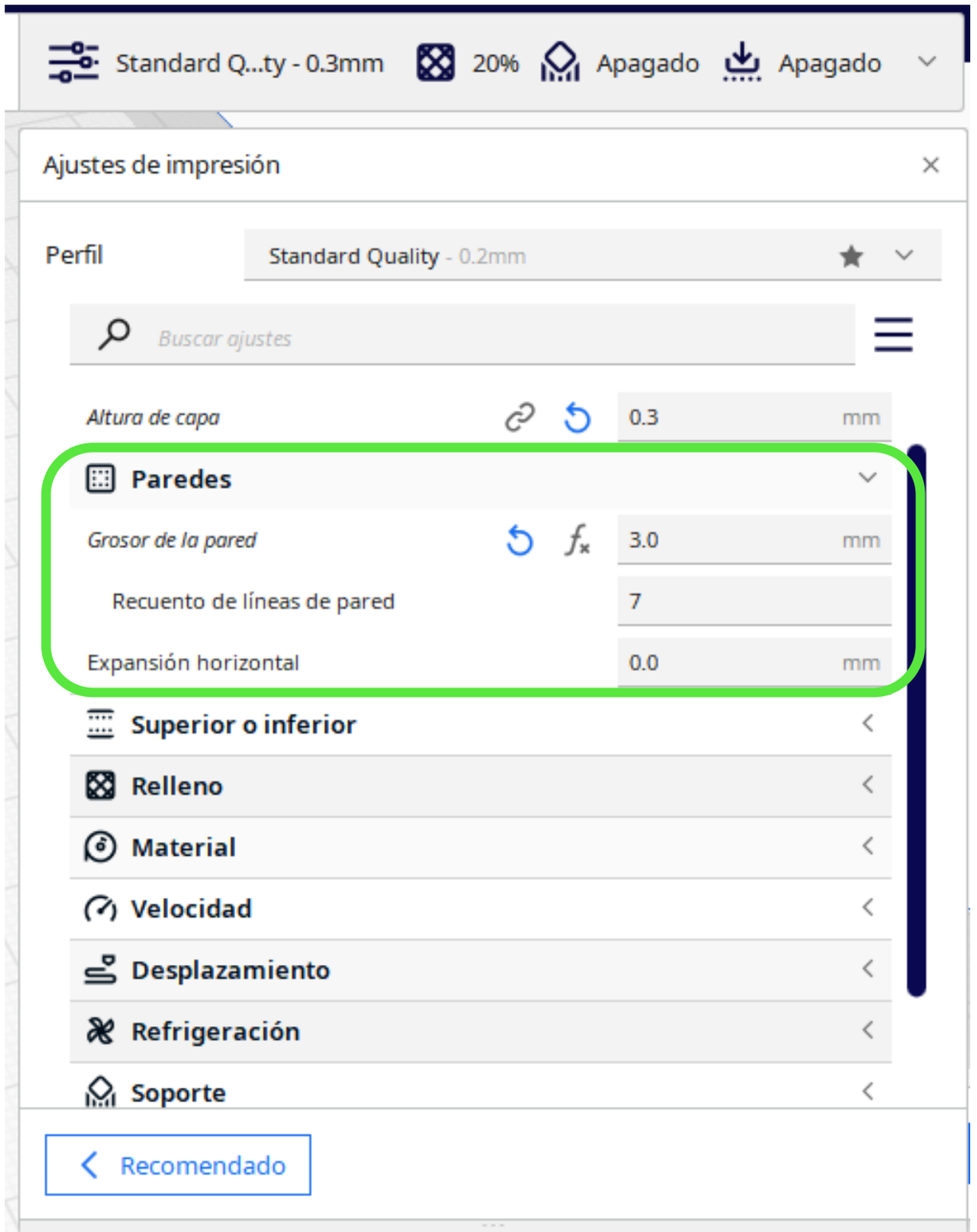
2 PAREDES

Paredes = 1



Paredes = 4

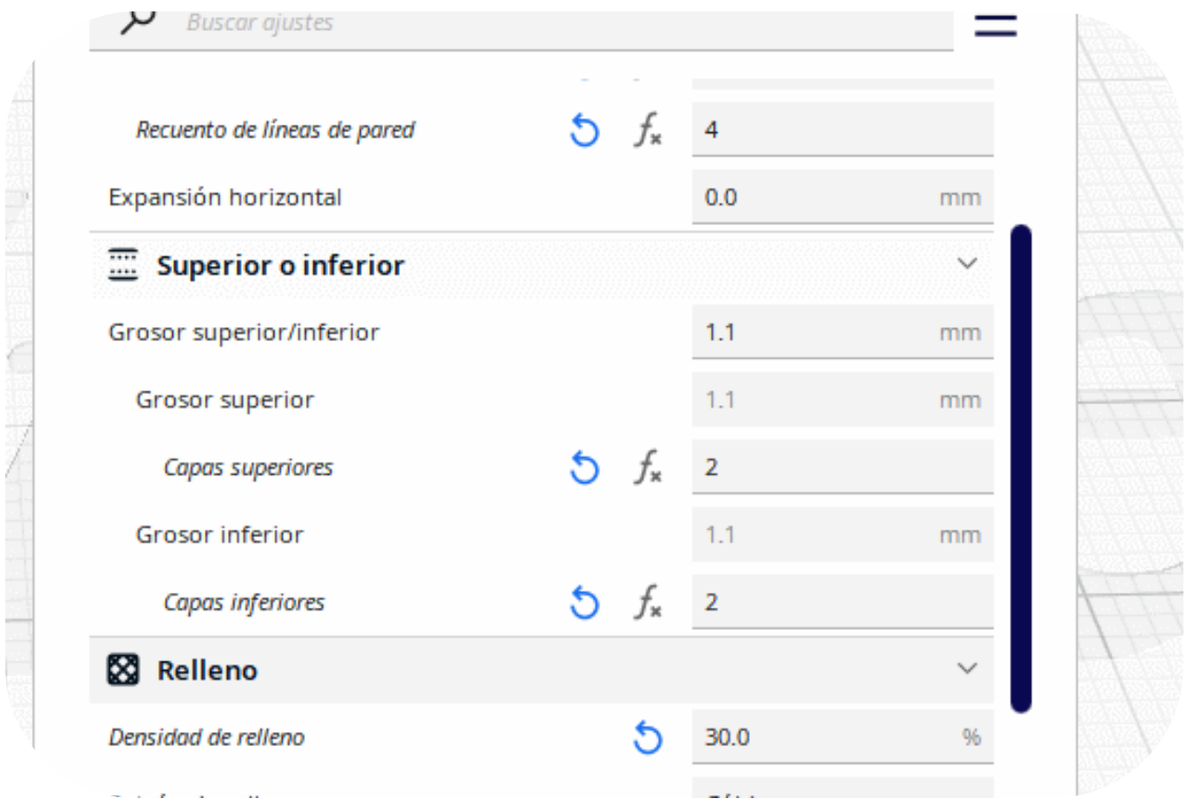


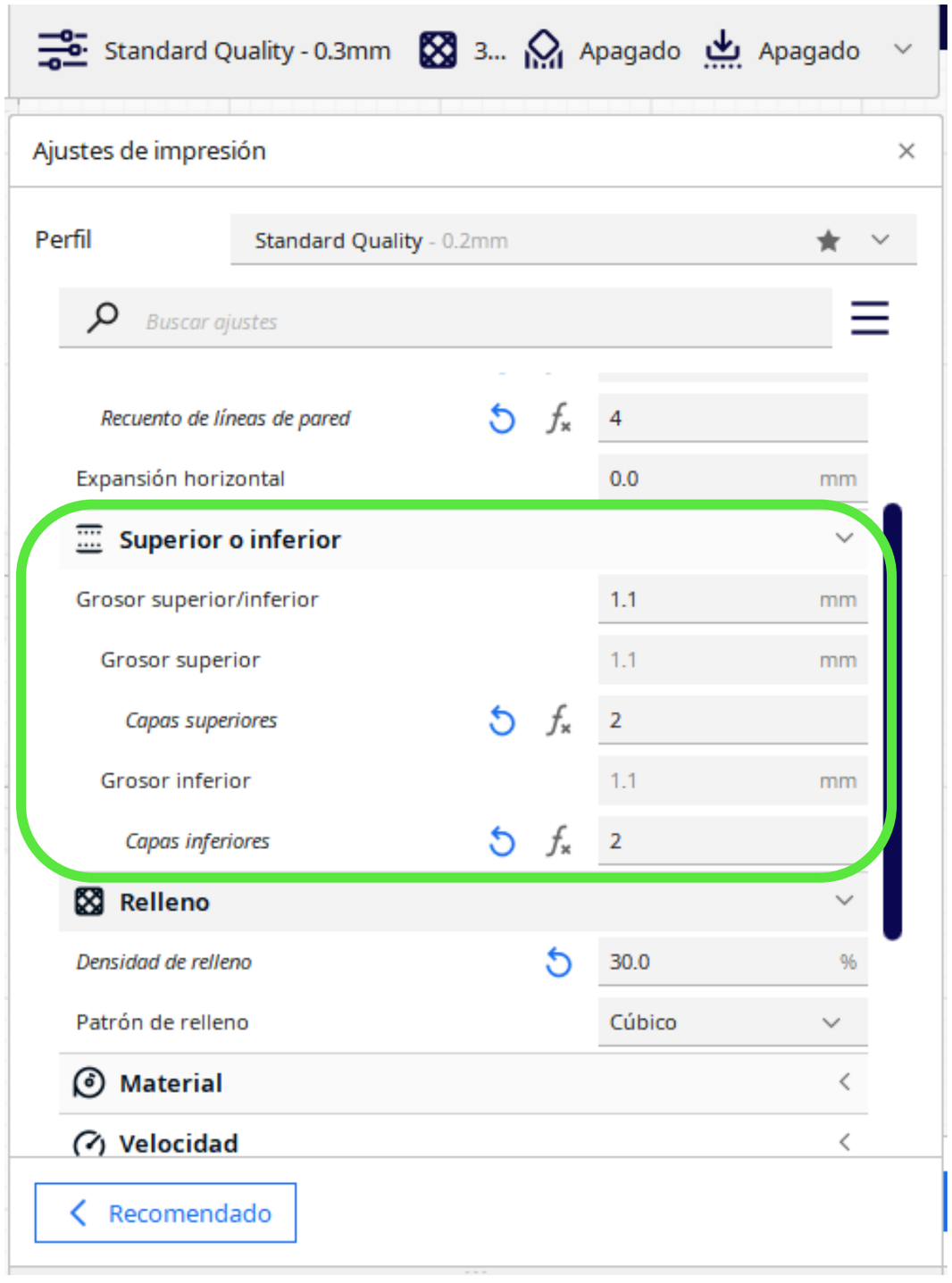


CURA– **Parámetros** de impresión

3 SUPERIOR E INFERIOR

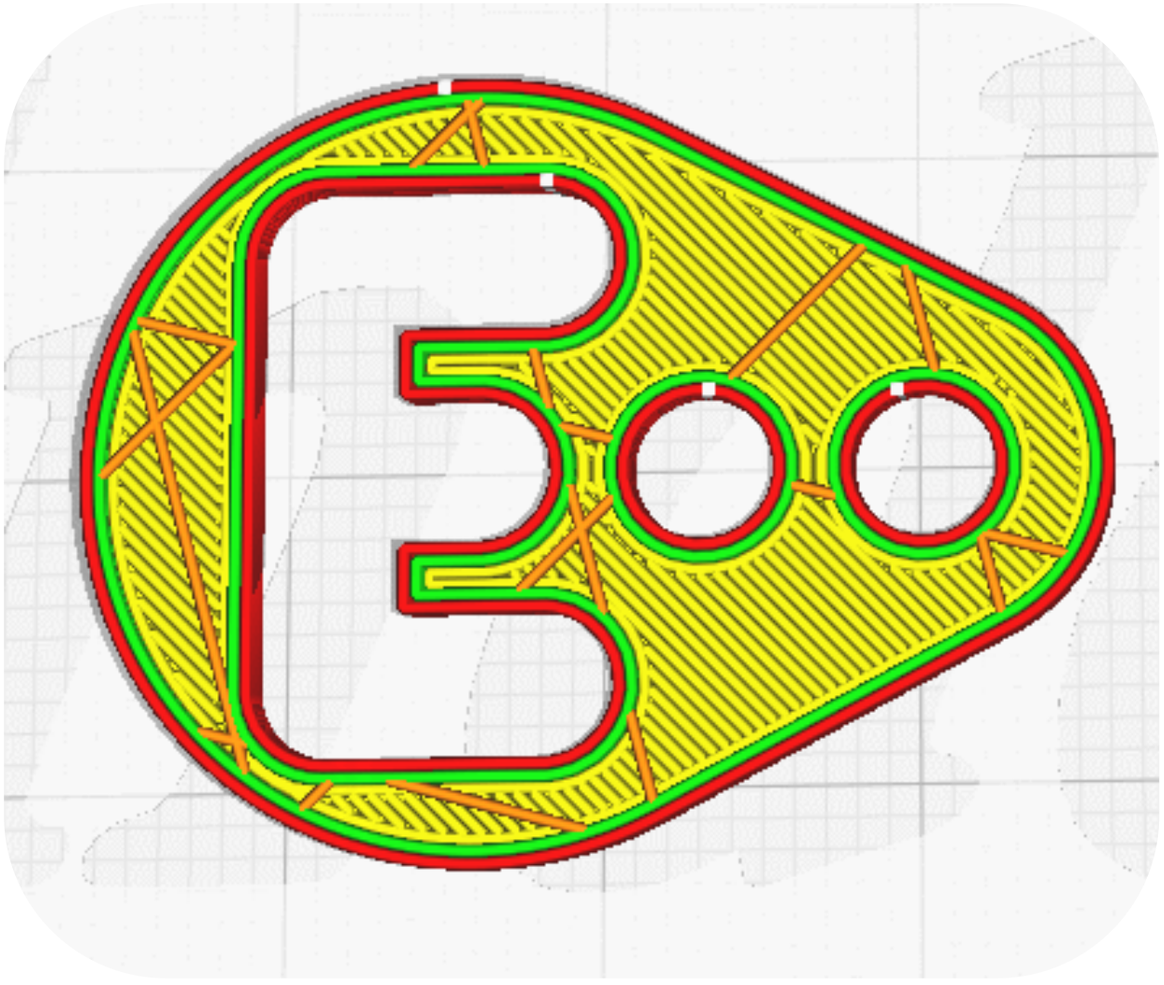
Capas Inferiores = 4



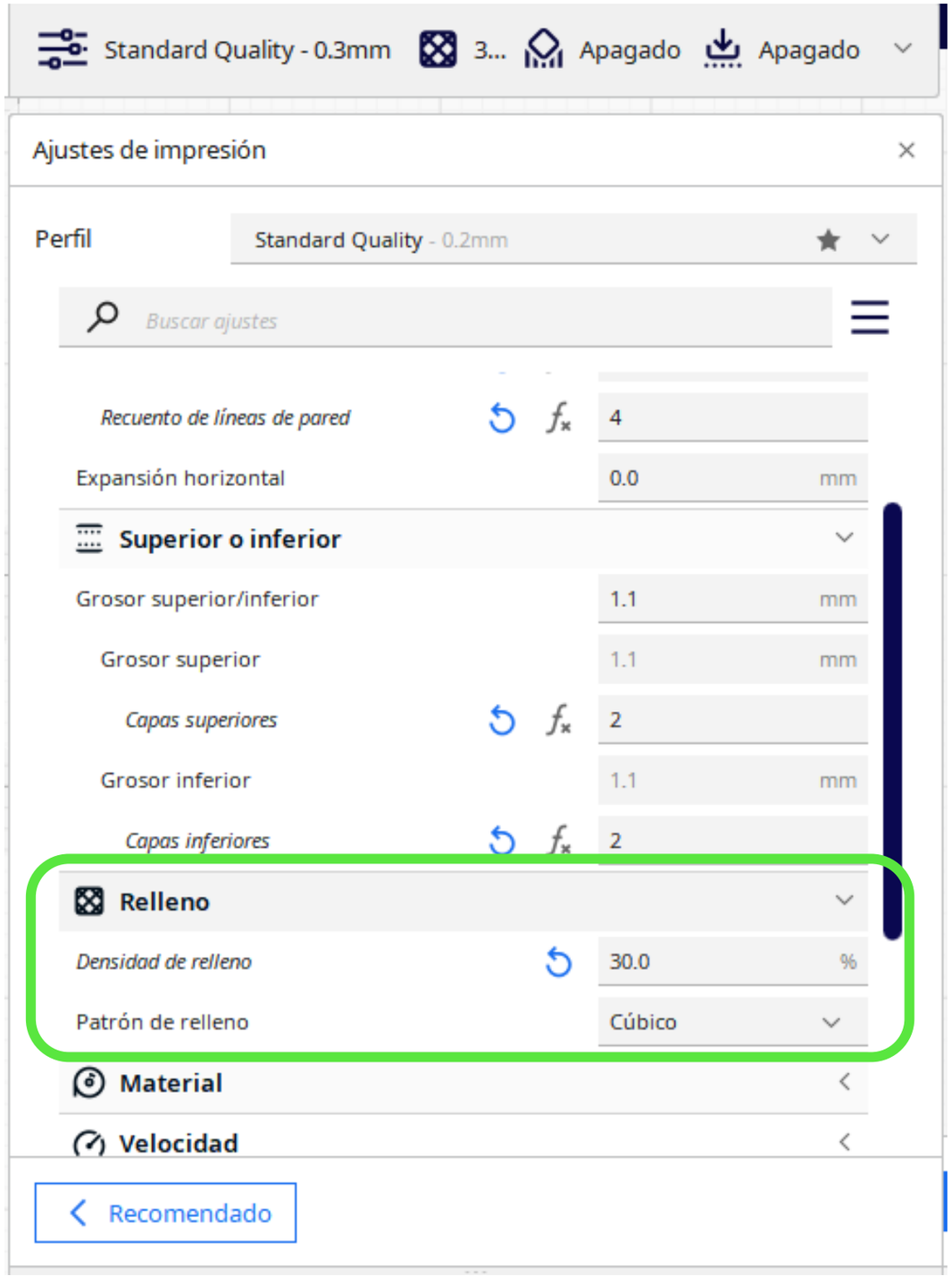


CURA– **Parámetros** de impresión

4 RELLENO



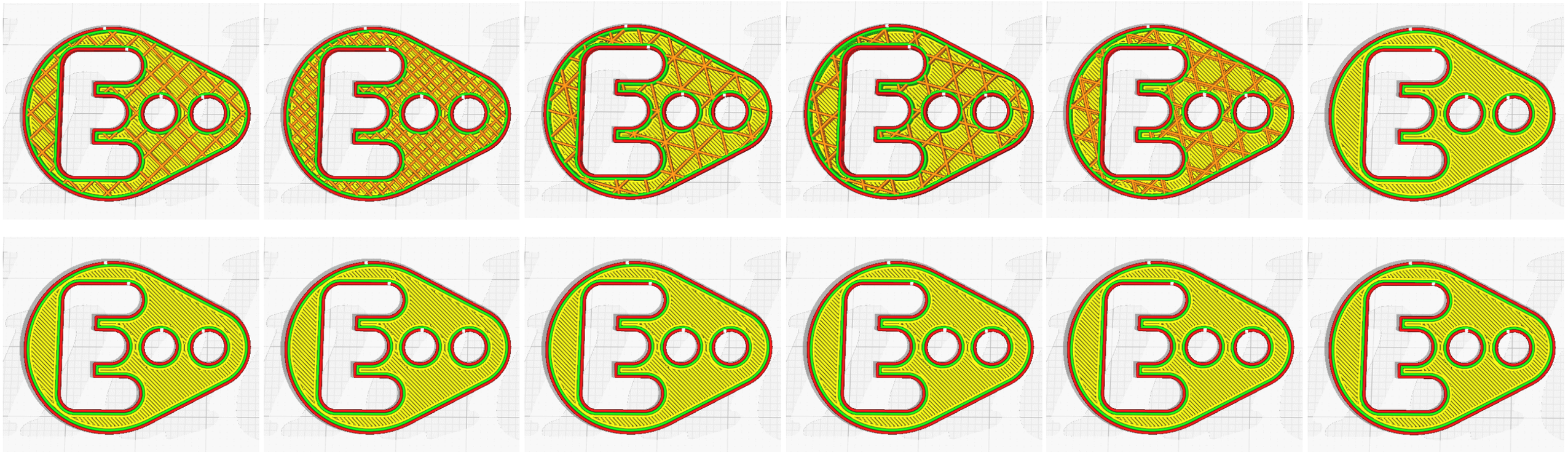
Densidad de relleno



CURA– **Parámetros** de impresión

4 RELLENO

Patrones



TESTING 3D printed INFILL PATTERNS for their STRENGTH

CURA– **Parámetros** de impresión

4 RELLENO

Patrones: recomendaciones

Característica	**Patrón de relleno**
Rapidez de impresión	Zig – Zag / Líneas / Iluminación
Menor gasto de filamento	Iluminación
Tenacidad	Giroide
Rigidez	Tri-hexagon / Cúbico
Flexibilidad	Cruz 3D
Calidad capa superior	Zig – Zag / Rejilla

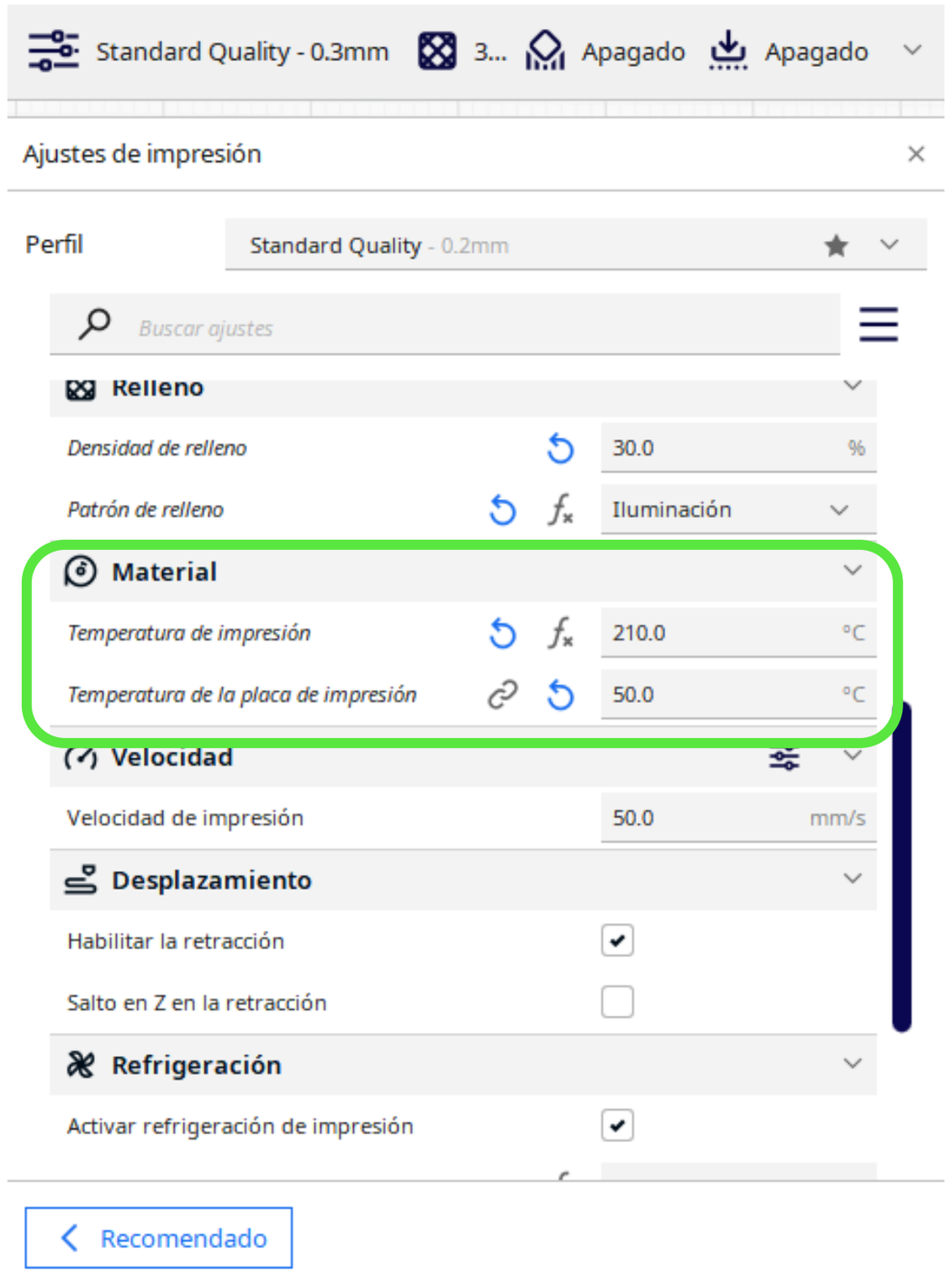
Cura Guide to the Best Infill Patterns

CURA– **Parámetros** de impresión

5 MATERIAL

PLA

190 – 220 ºC



CURA– **Parámetros** de impresión

5 MATERIAL

953 041 993 - Lunes - Viernes 08:00h a 14:00h y de 15:00h a 17:00h

smart materials 3D

Buscar

≡ TODAS LAS CATEGORÍAS | OFERTAS | CATÁLOGO | SMART CYCLE | RESINAS | 3D PRINTERS LAB | TUTORIALES | BLOG

Inicio / PLA / PLA GLITTER

PLA GLITTER

Lee las opiniones de los usuarios (19)

20,45 €



210 ºC

Descripción | Detalles del producto | Descargas

Reciclable	Reciclable
Apto contacto alimentos	Apto
Biocompostable	Biocompostable
Flexibilidad	Baja
Resistencia a impactos	Baja
Temperatura de impresión	210 ± 10ºC
Tipo de filamento PLA	GLITTER (Metalizado)

Referencia SMPLA0SV1A075
ean13 8434000012381

Standard Quality - 0.3mm | 3... | Apagado | Apagado

Ajustes de impresión

Perfil: Standard Quality - 0.2mm

Buscar ajustes

Relleno

Densidad de relleno 30.0 %

Patrón de relleno Iluminación

Material

Temperatura de impresión 210.0 °C

Temperatura de la placa de impresión 50.0 °C

Velocidad

Velocidad de impresión 50.0 mm/s

Desplazamiento

Habilitar la retracción ☑

Salto en Z en la retracción ☐

Refrigeración

Activar refrigeración de impresión ☑

Recomendado

CURA– **Parámetros** de impresión

5 MATERIAL

Temperatura
Cama Caliente
50 - 60 ºC

